Main Routine

EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface
EDC2 - v2.03 added Cycle Data Type
EDC3 - v3.00 added Quality RB03 and RB04, Scanner Tool updates
v3.50 - L980 Controller; add BulkFaultMsg Processing

0

S:FS

Program Version

aoiEDC_ProgVersion

aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE

Source	201
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD

Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Configuration

2

S:FS

Toggle ON to update Configuration ManualConfig

Toggle ON to update Configuration ManualConfig

U

Configuration

JSR

Routine Name	_211_Config

Initialize Line Loop Count to Service n Lines / Scan

3

CLR

Dest	LineLoopCount 9

Main Routine

Increment Line Loop Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

4

lbl_LineLoop
LBL

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line array Index
ADD
Source A LNi 27
Source B 1
Dest LNi 27

Line array Index
LT
Source A LNi 27
Source B CFG.LN.si <EDC.CFG.LN.si> 0

Line array Index
GT
Source A LNi 27
Source B CFG.LN.ei <EDC.CFG.LN.ei> 28

Line array Index
MOVE
Source CFG.LN.si <EDC.CFG.LN.si> 0
Dest LNi 27

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line array Index
MOVE
Source _ViewLNi 0
Dest LNi 27

Line Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

5

Line Loop for LNi
JSR
Routine Name _220_LineLoop

Increment Line Loop Count
Loop until all Max Lines serviced

service n Lines per Scan
set n to optimize EqDriver RxBuffer processing speed vs Gateway Scan & Class 3 Utilization

6

ADD
Source A LineLoopCount 9
Source B 1
Dest LineLoopCount 9

LT
Source A LineLoopCount 9
Source B CfgLineLoopMax 9

lbl_LineLoop
JMP

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

7

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	0
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	0
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	0
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

8

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

9

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
62
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.2
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

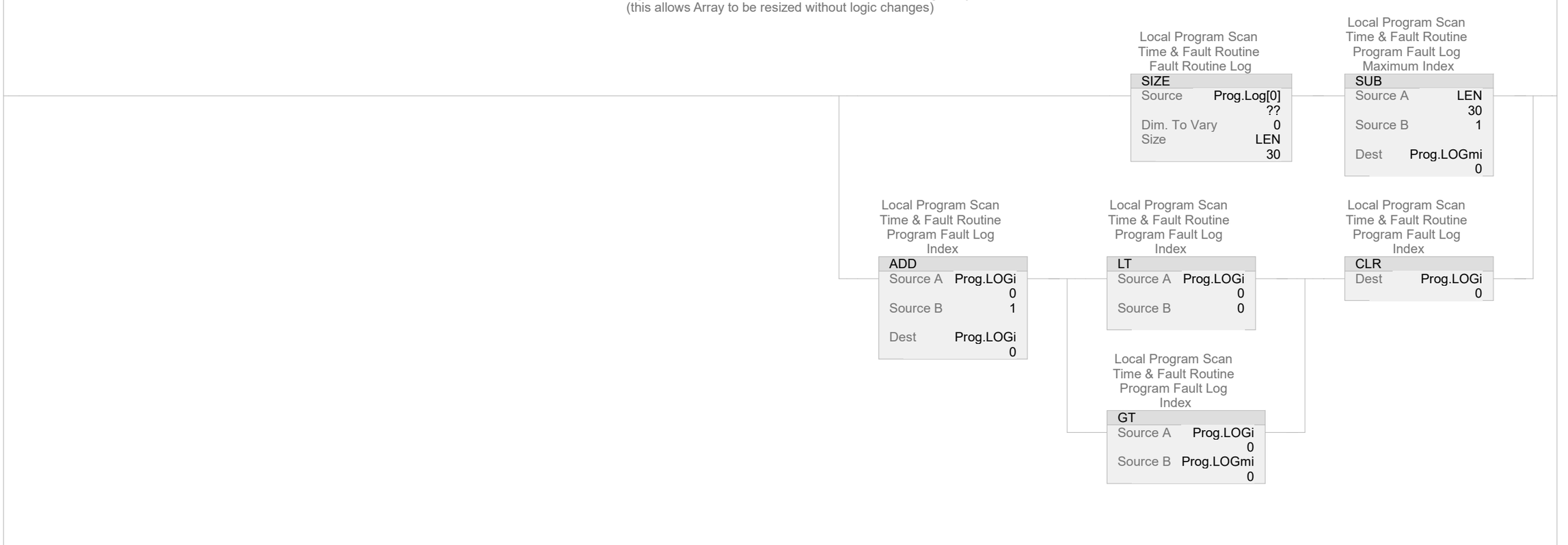
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

3

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

v1.19 Added CNVi, CARi, DATi

Local Program Scan Time & Fault Routine Fault Record

COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter

MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number

MOVE
Source Prog.Routine
225
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner

MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)

MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)

MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)

MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)

MOVE
Source LNi
27
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)

MOVE
Source EQi
26
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)

MOVE
Source CNVi
25
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)

MOVE
Source CARi
2
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)

MOVE
Source DATi
1
Dest Prog.Log[Prog.LOGi].Opt5
0

4

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 202
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Gateway Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 211
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

EquipRxBufferData Program - Initialize Line Loop Max to Service n Lines / Scan

Servicing more Lines/Scan allows EqDrivers to send data faster
The trade-off is an increase in Gateway scan time and Class 3 Message utiliation
(if you increase the Lines/Scan -- verify Gateway has the capacity!)

v3.5 use the Line array config to service ~750-1000 of 3000 Equipment/scan (target 3-4 scans for all Lines)

2

RxBufferData Quantity of Lines to service in one scan

DIV
Source A CFG.LN.n <EDC.CFG.LN.n> 29
Source B 3
Dest CfgLineLoopMax 9

Configuration

3

Bulk Fault Buffer Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

Equipment Data
Buffer Fault/Status
Receive Messages
(Header DINT, 10
Msgs 14 DINT/each,
Footer DINT

SIZE
Source EDC_EqRxBufferFault.Line[0].Equip[0].Fault[0]
0
Dim. To Vary 0
Size CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143

EDC Config Array Max
Index

SUB
Source A CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143
Source B 1
Dest CFG.BulkFaultBufferMSG.mi
<EDC.CFG.BulkFaultBufferMSG.mi>
142

EDC Config Array End
Count (udt size)

MOVE
Source 143
Dest CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143

EDC Config Array End
Count (udt size)

GT
Source A CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143
Source B CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143

EDC Config Array End
Count (udt size)

MOVE
Source CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143
Dest CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143

EDC Config Array End
Index (used for
copies)

SUB
Source A CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143
Source B 1
Dest CFG.BulkFaultBufferMSG.ei
<EDC.CFG.BulkFaultBufferMSG.ei>
142

Configuration

Bulk Fault Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

4

Equipment Data
Bulk Fault/Status
Message Equipment
Data Fault/Status
Message

SIZE	
Source	BulkFaultMsg.Msg[0] ??
Dim. To Vary	0
Size	CFG.BulkFaultMSG.n <EDC.CFG.BulkFaultMSG.n> 10

EDC Config Array Max
Index

SUB	
Source A	CFG.BulkFaultMSG.n <EDC.CFG.BulkFaultMSG.n> 10
Source B	1
Dest	CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9

Configuration

Fault Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

5

Equipment Data Buffer Fault/Status Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Fault[0]
1
Dim. To Vary 0
Size CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14

EDC Config Array Max Index

SUB
Source A CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14
Source B 1
Dest CFG.FaultMSG.mi <EDC.CFG.FaultMSG.mi> 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14
Source B CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14
Dest CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14
Source B 1
Dest CFG.FaultMSG.ei <EDC.CFG.FaultMSG.ei> 13

Configuration

Quality Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

6

Equipment Data Buffer Quality/Health Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Quality[0]
0
Dim. To Vary 0
Size CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39

EDC Config Array Max Index

SUB
Source A CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39
Source B 1
Dest CFG.QualityMSG.mi
<EDC.CFG.QualityMSG.mi>
38

EDC Config Array End Count (udt size)

MOVE
Source 39
Dest CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39

EDC Config Array End Count (udt size)

GT
Source A CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39
Source B CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39

EDC Config Array End Count (udt size)

MOVE
Source CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39
Dest CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39

EDC Config Array End Index (used for copies)

SUB
Source A CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39
Source B 1
Dest CFG.QualityMSG.ei
<EDC.CFG.QualityMSG.ei>
38

Configuration

Trend Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

7

Equipment Data Buffer Trend Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Trend[0]
0
Dim. To Vary 0
Size CFG.TrendMSG.n <EDC.CFG.TrendMSG.n>
70

EDC Config Array Max Index

SUB
Source A CFG.TrendMSG.n <EDC.CFG.TrendMSG.n>
70
Source B 1
Dest CFG.TrendMSG.mi <EDC.CFG.TrendMSG.mi>
69

EDC Config Array End Count (udt size)

MOVE
Source 70
Dest CFG.TrendMSG.e <EDC.CFG.TrendMSG.e>
70

EDC Config Array End Count (udt size)

GT
Source A CFG.TrendMSG.e <EDC.CFG.TrendMSG.e>
70
Source B CFG.TrendMSG.n <EDC.CFG.TrendMSG.n>
70

EDC Config Array End Count (udt size)

MOVE
Source CFG.TrendMSG.n <EDC.CFG.TrendMSG.n>
70
Dest CFG.TrendMSG.e <EDC.CFG.TrendMSG.e>
70

EDC Config Array End Index (used for copies)

SUB
Source A CFG.TrendMSG.e <EDC.CFG.TrendMSG.e>
70
Source B 1
Dest CFG.TrendMSG.ei <EDC.CFG.TrendMSG.ei>
69

Configuration

8

Version Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

Equipment Data Buffer Version Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Version[0]
8
Dim. To Vary 0
Size CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30

EDC Config Array Max Index

SUB
Source A CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30
Source B 1
Dest CFG.VersionMSG.mi <EDC.CFG.VersionMSG.mi>
29

EDC Config Array End Count (udt size)

MOVE
Source 30
Dest CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30

EDC Config Array End Count (udt size)

GT
Source A CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30
Source B CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30

EDC Config Array End Count (udt size)

MOVE
Source CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30
Dest CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30

EDC Config Array End Index (used for copies)

SUB
Source A CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30
Source B 1
Dest CFG.VersionMSG.ei <EDC.CFG.VersionMSG.ei>
29

Configuration

Cycle Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

9

Equipment Data Buffer Cycle Receive Message

SIZE
Source EDC_EqRxBufferAux.Line[0].Equip[0].Cycle[0] 0
Dim. To Vary 0
Size CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134

EDC Config Array Max Index

SUB
Source A CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134
Source B 1
Dest CFG.CycleMSG.mi <EDC.CFG.CycleMSG.mi> 133

EDC Config Array End Count (udt size)

MOVE
Source 134
Dest CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134

EDC Config Array End Count (udt size)

GT
Source A CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134
Source B CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134

EDC Config Array End Count (udt size)

MOVE
Source CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134
Dest CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134
Source B 1
Dest CFG.CycleMSG.ei <EDC.CFG.CycleMSG.ei> 133

(End)

Line Loop for LNi

Line Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	220
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

EDC Config Array Indexes for Line-Equip

COP	
Source	EDC.Line[LNi].CFG.EQ
Dest	LineCfgEQ
Length	1

Line Loop for LNi

Equipment Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

3



Copy Line working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

only Line.CFG used - no changes to save

4



Exit Routine

5



Line Loop for LNi

Check Viewing Index Limits

6

Line View Tag Index
LT
Source A _ViewLNi 0
Source B 0

Line View Tag Index
GT
Source A _ViewLNi 0
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Line View Tag Index
CLR
Dest _ViewLNi 0

Copy Line Instance to Viewing Tag if selected
(_ViewLine can be used to multiplex Trends and Diagnostic Spreadsheets)

7

_ViewEN

Line array Index
EQ
Source A LNi 27
Source B _ViewLNi 0

Line View Tag (used by View Diagnostic Spreadsheet)
COP
Source EDC.Line[LNi]
Dest _ViewLine
Length 1

(End)

Equipment Loop for EQi

Equipment Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 225
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Exit if Equipment is not Configured
(reduce scan time by skipping logic for unused Equipment)
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)
EDC2 - Nov 15, 2022 - disable/ignore EQID 0000

2

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Config

lbl_Exit
JMP

Line array Index

EQ
Source A LNi 27
Source B 0

Equipment array Index

EQ
Source A EQi 26
Source B 0

Equipment Loop for EQi

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

3

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	Equip
Length	1

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDCAux.Line[LNi].Equip[EQi]
Dest	EquipAux
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi

COP	
Source	EDC_Status01.Line[LNi].Equip[EQi]
Dest	EquipStatus01
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status02.Line[LNi].Equip[EQi]
Dest	EquipStatus02
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status03.Line[LNi].Equip[EQi]
Dest	EquipStatus03
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status04.Line[LNi].Equip[EQi]
Dest	EquipStatus04
Length	1

Equipment Comm Status
-- Set Default State = Configured if no State value
(don't run Equipment Timer)

4

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP	
Source	EquipStatus01.State
Dest	EquipStatus01.StateVal
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

LE	
Source A	EquipStatus01.StateVal
	16#0000_0022
Source B	1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State

FLL	
Source	0
Dest	EquipStatus01.State
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi State = White - Configured (never received any Data Requests)

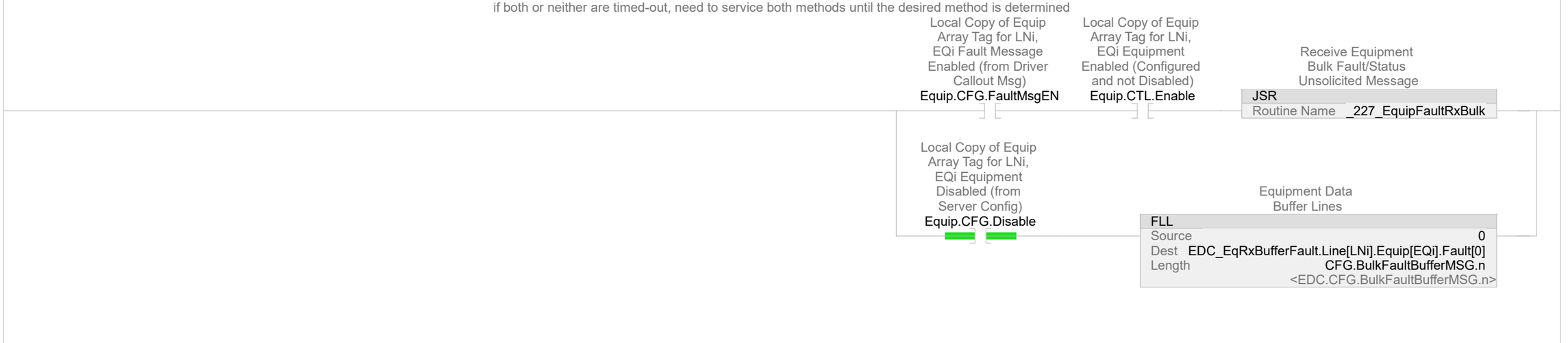
EquipStatus01.State.CFG (L)

Equipment Loop for EQi

Bulk Fault Event Message
DHF006 - if disabled, block FaultMsg
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

Fault messages for each Equipment should only arrive on EDC_EqRxBuffer (single) or EDC_EqRxBufferFault (bulk)
normally, one method will timeout while the other operates
if both or neither are timed-out, need to service both methods until the desired method is determined

5

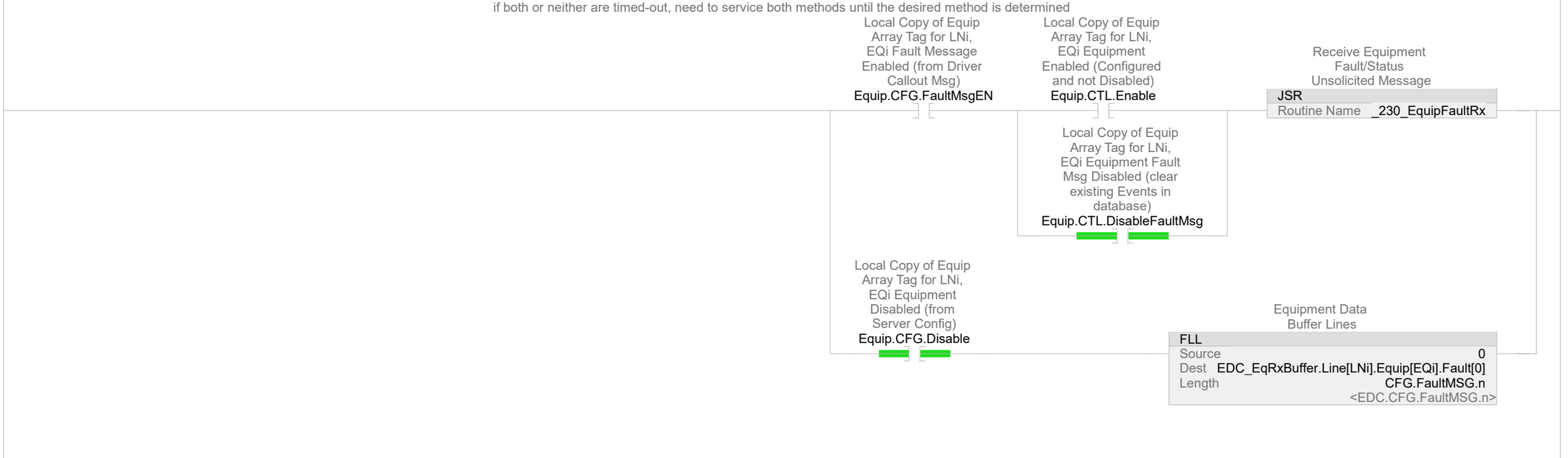


Equipment Loop for EQi

Fault Event Message
DHF006 - if disabled, block FaultMsg after sending PKT 0-7 Messages to reset all Fault bits (addition of Disabled Equipment support)
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

Fault messages for each Equipment should only arrive on EDC_EqRxBuffer (single) or EDC_EqRxBufferFault (bulk)
normally, one method will timeout while the other operates
if both or neither are timed-out, need to service both methods until the desired method is determined

6



Equipment Loop for EQi

Quality/Health Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

7

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Quality/Health Unsolicited Message

JSR
Routine Name _240_EquipQualityRx

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Length CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>

Equipment Loop for EQi

Trend Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL18 - allow PdM Sample for Equipment without Trend Message enabled
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

8

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Trend Unsolicited Message

JSR
Routine Name _250_EquipTrendRx

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Length CFG.TrendMSG.n
<EDC.CFG.TrendMSG.n>

Version Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

9

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Version Unsolicited Message

JSR
Routine Name _260_EquipVersionRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Length CFG.VersionMSG.n
<EDC.CFG.VersionMSG.n>

Equipment Loop for EQi

Cycle Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - when Disabled, discard Messages from RxBuffer
EDC2 WL08 - add Cycle Data Message

10

Local Copy of Equip Array Tag for LNi, EQi Cycle Message Enabled (from Driver Callout Msg)
Equip.CFG.CycleMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Cycle Unsolicited Message
JSR
Routine Name _270_EquipCycleRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Length CFG.CycleMSG.n
<EDC.CFG.CycleMSG.n>

Save & Exit Routine

11

lbl_SaveExit
LBL

NOP

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

Equipment Data Collection Lines and Equipment Array
COP
Source Equip
Dest EDC.Line[LNi].Equip[EQi]
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source EquipAux
Dest EDCAux.Line[LNi].Equip[EQi]
Length 1

Server Interface Status Lines and Equipment Array
COP
Source EquipStatus01
Dest EDC_Status01.Line[LNi].Equip[EQi]
Length 1

Event Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus02
Dest EDC_Status02.Line[LNi].Equip[EQi]
Length 1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus03
Dest EDC_Status03.Line[LNi].Equip[EQi]
Length 1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus04
Dest EDC_Status04.Line[LNi].Equip[EQi]
Length 1

Equipment Loop for EQi

Exit Routine

13

lbl_Exit
LBL

NOP

Check Viewing Index Limits

14

Equipment View Tag Index
LT
Source A _ViewEQi 6
Source B 0

Equipment View Tag Index
GT
Source A _ViewEQi 6
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Equipment View Tag Index
CLR
Dest _ViewEQi 6

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

15

_ViewEN

Line array Index
EQ
Source A LNi 27
Source B _ViewLNi 0

Equipment array Index
EQ
Source A EQi 26
Source B _ViewEQi 6

Equipment View Tag (used by View Diagnostic Spreadsheet)
COP
Source EDC.Line[LNi].Equip[EQi]
Dest _ViewEquip
Length 1

(End)

Receive Equipment Bulk Fault/Status Unsolicited Message

Receive Equipment Bulk Fault/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 227
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Bulk Fault/Status Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

Transaction Timeout if both Rx & RxBulk timeout
DHF006 - Block Reset for Send Disable Message

2

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest EquipAux.CTL.BulkFaultMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer EquipAux.CTL.BulkFaultMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout

EquipAux.CTL.BulkFaultMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2004
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A BulkFaultTimeoutER
13539
Source B 1
Dest BulkFaultTimeoutER
13539

Equipment Data Bulk Fault/Status Message

FLL
Source 0
Dest BulkFaultMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

Bulk Fault Message contains 1-10 Fault Messages

3

Equipment Data
Bulk Fault/Status
Message

CPS
Source EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[0]
Dest BulkFaultMsg
Length 1

the message has a variable length depending on the number of Records
use the Record Count to update the Footer.TransNum
14 DINT/record + 2 DINT Header

4

Equipment Data
Bulk Fault/Status
Message Header -
Record Count

LIMIT
Low Limit 1
Test BulkFaultMsg.Header.RecordCount
0
High Limit 10

CPT
Dest FLTi
16
Expression (BulkFaultMsg.Header.RecordCount * CFG.FaultMSG.e) + 2

Equipment Data
Bulk Fault/Status
Message Footer

COP
Source EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[FLTi]
Dest BulkFaultMsg.Footer
Length 1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data
Bulk Fault/Status
Message Header -
Transaction Number
(must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data
Bulk Fault/Status
Message Footer -
Transaction Number
(must match Footer)

EQ
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing for Bulk & Legacy path

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest EquipAux.CTL.BulkFaultMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2000
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A: BulkFaultMsg.Header.TransNum 0
Source B: 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A: BulkFaultMsg.Footer.TransNum 0
Source B: 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source: CFG.PartialMsgTimeout <EDC.CFG.PartialMsgTimeout> 10000
Dest: EquipAux.CTL.BulkFaultMsg.PartialTMR.PRE 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer: EquipAux.CTL.BulkFaultMsg.PartialTMR
Preset: 0
Accum: 0
EN
DN

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A: BulkFaultMsg.Header.TransNum 0
Source B: 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A: BulkFaultMsg.Footer.TransNum 0
Source B: 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate

aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	BulkFaultMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	EquipAux.CTL.BulkFaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	2005
Input Par	EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	BulkFaultMsg.Header.TransNum

Diagnostic

ADD

Source A	BulkFaultPartialMsgER 272
Source B	1
Dest	BulkFaultPartialMsgER 272

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum
0
Source B BulkFaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

EquipAux.CTL.BulkFaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2006
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Diagnostic

ADD
Source A FaultTransNumER
0
Source B 1
Dest FaultTransNumER
0

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum 0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Footer.TransNum 0
Source B 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum 0
Source B BulkFaultMsg.Footer.TransNum 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2001
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

skip over the Message Processing Loop if a Transaction was not received

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
/

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

BulkFaultMsg.Header.EQID should match each FaultMsg.Header.EQID
(not going to check FaultMsg.Header.EQID for each Message - assume they are the same)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Expected Equipment ID
CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Bulk Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)
NE
Source A BulkFaultMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Equipment ID Mismatch ER
EquipStatus01.Update.FaultRxEqIdER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source BulkFaultMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2007
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Diagnostic
ADD
Source A BulkFaultEqidER 0
Source B 1
Dest BulkFaultEqidER 0

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

BulkFaultMsg.Header.TransNum should be incremented by the previous Message Record Count
(not going to check FaultMsg.Header.TransNum for each Message - assume they are in order)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum
0
Source B EquipAux.CTL.BulkFaultMsg.TransNumLast
0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EquipAux.CTL.BulkFaultMsg.TransNumLast
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.KA

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2002
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

BulkFaultMsg.Header.TransNum should match BulkFaultMsg.Msg[0].Header.TransNum (same as first Record)
(not going to check FaultMsg.Header.TransNum for each Message - assume they are in order)

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum 0
Source B EquipAux.CTL.BulkFaultMsg.TransNumLast 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum 0
Source B EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Transaction Number Sequence ER
EquipStatus01.Update.FaultRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2009
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Diagnostic

ADD
Source A BulkFaultSeqER 37
Source B 1
Dest BulkFaultSeqER 37

Diagnostic

MOVE
Source LNi 27
Dest BulkFaultSeqER_LNi 10

Diagnostic

MOVE
Source EQi 26
Dest BulkFaultSeqER_EQi 4

Diagnostic

MOVE
Source BulkFaultMsg.Header.TransNum 0
Dest BulkFaultSeqER_TransNum 1

Diagnostic

MOVE
Source EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Dest BulkFaultSeqER_TransNumNext 42

Receive Equipment Bulk Fault/Status Unsolicited Message

Initialize Record Loop Indexes
(if invalid, assume max count and process the blanks - hopefully the sequence errors will draw attention)

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Record Index
CLR
Dest RECi 2

Record Max Index
SUB
Source A BulkFaultMsg.Header.RecordCount 0
Source B 1
Dest RECmi 1

Record Max Index
LT
Source A RECmi 1
Source B 0

Record Max Index
GT
Source A RECmi 1
Source B CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9

Record Max Index
MOVE
Source CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9
Dest RECmi 1

Message Loop
Load Next FaultMsg

17

lbl_MsgLoop
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Fault/Status Message
COP
Source BulkFaultMsg.Msg[RECi]
Dest FaultMsg
Length 1

TBD -- should each FaultMsg be validated for EQID and TransNum?
it's a reasonable assumption the standard EqDrv properly packages multiple FaultMsg for the BulkMsg

EQID Check
Valid TransNum Check
Duplicate TransNum Check
TransNum Sequence Check

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

NOP

Receive Equipment Bulk Fault/Status Unsolicited Message

Load new message in Ring Buffer
Feb 2017 - 4 RB Transactions
v1.6 - 6 RB Transactions
Mar 5, 2018 - rebalance Ring Buffers to avoid backlogs in RB4 and RB6
Sep 2023 - change RB channel allocation to modulo of Line Number (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers
Dec 4, 2025 - change RB channel allocation back to modulo of LNi -- all EQID on same Line must go thru same RB to avoid incorrect Candidate/Event generation for Maximo/PCR

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

RingBuffer Modulo Result
MOD
Source A LNi 27
Source B 6
Dest BufferID 3

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 1

UseFaultRB01

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _231_EquipFaultRxLoadRB01

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 2

UseFaultRB02

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _232_EquipFaultRxLoadRB02

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 3

UseFaultRB03

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _233_EquipFaultRxLoadRB03

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 4

UseFaultRB04

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _234_EquipFaultRxLoadRB04

RingBuffer Modulo Result
EQ

UseFaultRB05

Load Equipment Fault/Status Message into Ring Buffer
JSR

Receive Equipment Bulk Fault/Status Unsolicited Message

EQ
Source A BufferID
3
Source B 5

UseFaultRB05

JSR
Routine Name _235_EquipFaultRxLoadRB05

Load Equipment
Fault/Status Message
into Ring Buffer

UseFaultRB01 UseFaultRB02 UseFaultRB03 UseFaultRB04 UseFaultRB05 UseFaultRB06

JSR
Routine Name _236_EquipFaultRxLoadRB06

Increment Record Index and jump to beginning of Record Loop

20

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Record
Index
ADD
Source A RECi
2
Source B 1
Dest RECi
2

Record
Index
LE
Source A RECi
2
Source B RECmi
1

lbl_MsgLoop
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Prepare for next message
TransNumNext is incremented by the Record Count

21

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source BulkFaultMsg.Header.TransNum 0
Dest EquipAux.CTL.BulkFaultMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A BulkFaultMsg.Header.TransNum 0
Source B BulkFaultMsg.Header.RecordCount 0
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

SUB
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Receive Equipment Bulk Fault/Status Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
DHF006 - add DisableFaultMsg

BulkFaultMsg - store last FaultMsg in EDC_EqRxData
(only one of Fault and BulkFault RxBuffer will be receiving data, so no problem sharing result area)

don't reset BulkFaultMsg Timeout if FaultMsg is in use
(timeout is used to suspend scan of the unused Fault method)

22

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset

Diagnostic
ADD
Source A BulkTransReset 272
Source B 1
Dest BulkTransReset 272

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Diagnostic
ADD
Source A BulkTransRcvd 27916223
Source B 1
Dest BulkTransRcvd 27916223

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
EquipAux.CTL.BulkFaultMsg.Timeout

Diagnostic
ADD
Source A BulkTransTimeout 20308
Source B 1
Dest BulkTransTimeout 20308

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
EquipAux.CTL.BulkFaultMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest EquipAux.CTL.BulkFaultMsg.TransLast 0

Equipment Data Lines
COP
Source FaultMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Fault
Length 1

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[0]
Length CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>

Exit Routine

23

lbl_Exit
LBL

NOP

(End)

Receive Equipment Fault/Status Unsolicited Message

Receive Equipment Fault/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 230
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Fault/Status Unsolicited Message

Send Fault Event message for PKTi 0-7 with all Events off to reset any open Events.

2

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
EQ
Source A Equip.CTL.FaultMsg.TransNumNext 17
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
MOVE
Source 1
Dest Equip.CTL.FaultMsg.TransNumNext 17

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)
MOVE
Source Equip.CTL.FaultMsg.TransNumNext 17
Dest FaultMsg.Header.TransNum 0

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)
CPT
Dest FaultMsg.Header.EQID 0
Expression (LNi * 100) + EQi

Equipment Data Fault/Status Message Header - Data Count - Total Qty of Words used in all Packets
MOVE
Source 8
Dest FaultMsg.Header.DataCount 0

Equipment Data Fault/Status Message Header - Packet ID - generates offset for 8-DINT mulitplexed data packet
MOVE
Source Equip.CTL.DisablePKTi 0
Dest FaultMsg.Header.PacketID 0

Equipment Data Fault/Status Message Transaction Time Stamp
COP
Source CLX.TransTime <EDC.CLX.TransTime>
Dest FaultMsg.TransTime
Length 1

Equipment Data Fault/Status Message Data - Predefined Status
FLL
Source 0
Dest FaultMsg.PreDef
Length 1

Equipment Data Fault/Status Message Data - Equipment Defined Status
FLL
Source 0
Dest FaultMsg.EqDef
Length 1

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)
MOVE
Source FaultMsg.Header.TransNum 0
Dest FaultMsg.Footer.TransNum 0

Receive Equipment Fault/Status Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

Transaction Timeout if both Rx & RxBulk timeout
DHF006 - Block Reset for Send Disable Message

3

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
/

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.FaultMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.FaultMsg.TransTMR
Preset 999999999
Accum 44481
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.FaultMsg.TransTMR.ACC 44481
Source B CFG.BufferMsgTimeout <EDC.CFG.BufferMsgTimeout> 330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2004
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A FaultTimeoutER 20453
Source B 1
Dest FaultTimeoutER 20453

Equipment Data Fault/Status Message

FLL
Source 0
Dest FaultMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT
DHF006 - Block Buffer Data copy for Send Disable Message

4

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
[/]

Equipment Data Fault/Status Message

CPS	
Source	EDC_EqRxBuffer.Line[LNi].Equip[EQi].Fault[0]
Dest	FaultMsg
Length	1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ	
Source A	FaultMsg.Header.TransNum
	0
Source B	0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

EQ	
Source A	FaultMsg.Footer.TransNum
	0
Source B	0

lbl_Exit
JMP

Receive Equipment Fault/Status Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing for Legacy & Bulk path

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.FaultMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2000
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Receive Equipment Fault/Status Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.FaultMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.FaultMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Fault/Status Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.FaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2005
Input Par	Equip.CTL.FaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	FaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	FaultPartialMsgER 0
Source B	1
Dest	FaultPartialMsgER 0

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B FaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.FaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2006
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Diagnostic

ADD
Source A FaultTransNumER
0
Source B 1
Dest FaultTransNumER
0

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Header.TransNum
0
Source B FaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2001
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Receive Equipment Fault/Status Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 1006
Expression	LNi*100+EQi

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	FaultMsg.Header.EQID 0
Source B	ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Equipment ID Mismatch ER
EquipStatus01.Update.FaultRxEqIdER
(L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultMsg.Header.EQID 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.FaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2007
Input Par	Equip.CTL.FaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	FaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	FaultEqidER 56723
Source B	1
Dest	FaultEqidER 56723

lbl_MsgReset
(JMP)

Receive Equipment Fault/Status Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A: FaultMsg.Header.TransNum
0
Source B: Equip.CTL.FaultMsg.TransNumLast
16

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.FaultMsg.TransNumLast
16
Dest: Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.FaultMsg
PerfTrans: EquipStatus02.FaultRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2002
Input Par: Equip.CTL.FaultMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: FaultMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	FaultMsg.Header.TransNum 0
Source B	Equip.CTL.FaultMsg.TransNumLast 16

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	FaultMsg.Header.TransNum 0
Source B	Equip.CTL.FaultMsg.TransNumNext 17

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Transaction Number Sequence ER
EquipStatus01.Update.FaultRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.FaultMsg.TransNumNext 17
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.FaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2009
Input Par	Equip.CTL.FaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	FaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	FaultSeqER 1817812
Source B	1
Dest	FaultSeqER 1817812

Diagnostic

MOVE	
Source	LNi 27
Dest	FaultSeqER_LNi 1

Diagnostic

MOVE	
Source	EQi 26
Dest	FaultSeqER_EQi 19

Diagnostic

MOVE	
Source	FaultMsg.Header.TransNum 0
Dest	FaultSeqER_TransNum 6136

Diagnostic

MOVE	
Source	Equip.CTL.FaultMsg.TransNumNext 17
Dest	FaultSeqER_TransNumNext 6108

Receive Equipment Fault/Status Unsolicited Message

Receive Equipment Fault/Status Unsolicited Message

Load new message in Ring Buffer
Feb 2017 - 4 RB Transactions
v1.6 - 6 RB Transactions
Mar 5, 2018 - rebalance Ring Buffers to avoid backlogs in RB4 and RB6
Sep 2023 - change RB channel allocation to modulo of Line Number (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers
Dec 4, 2025 - change RB channel allocation back to modulo of LNi -- all EQID on same Line must go thru same RB to avoid incorrect Candidate/Event generation for Maximo/PCR

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

RingBuffer Modulo Result
MOD
Source A LNi 27
Source B 6
Dest BufferID 3

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 1

UseFaultRB01

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _231_EquipFaultRxLoadRB01

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 2

UseFaultRB02

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _232_EquipFaultRxLoadRB02

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 3

UseFaultRB03

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _233_EquipFaultRxLoadRB03

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 4

UseFaultRB04

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _234_EquipFaultRxLoadRB04

RingBuffer Modulo Result
EQ

UseFaultRB05

Load Equipment Fault/Status Message into Ring Buffer
JSR

Receive Equipment Fault/Status Unsolicited Message

EQ
Source A BufferID
3
Source B 5

UseFaultRB05

JSR
Routine Name _235_EquipFaultRxLoadRB05

Load Equipment
Fault/Status Message
into Ring Buffer

UseFaultRB01 UseFaultRB02 UseFaultRB03 UseFaultRB04 UseFaultRB05 UseFaultRB06

JSR
Routine Name _236_EquipFaultRxLoadRB06

Prepare for next message

16

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Received
Equip.CTL.FaultMsg.TransRcvd

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Last Transaction
Number

MOVE
Source FaultMsg.Header.TransNum
0
Dest Equip.CTL.FaultMsg.TransNumLast
16

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

ADD
Source A FaultMsg.Header.TransNum
0
Source B 1
Dest Equip.CTL.FaultMsg.TransNumNext
17

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

LT
Source A Equip.CTL.FaultMsg.TransNumNext
17
Source B 1

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

GT
Source A Equip.CTL.FaultMsg.TransNumNext
17
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

MOVE
Source 1
Dest Equip.CTL.FaultMsg.TransNumNext
17

Receive Equipment Fault/Status Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
DHF006 - add DisableFaultMsg

don't reset FaultMsg Timeout if BulkFaultMessage is in use
(timeout is used to suspend scan of the unused Fault method)

17

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset

Diagnostic
ADD
Source A TransReset 0
Source B 1
Dest TransReset 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Diagnostic
ADD
Source A TransRcvd 59530117
Source B 1
Dest TransRcvd 59530117

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout

Diagnostic
ADD
Source A TransTimeout 13563
Source B 1
Dest TransTimeout 13563

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Diagnostic
ADD
Source A DisableFaultMsg 192
Source B 1
Dest DisableFaultMsg 192

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.FaultMsg.TransLast 0

Equipment Data Lines
COP
Source FaultMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Fault
Length 1

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Fault[0]
Length CFG.FaultMSG.n <EDC.CFG.FaultMSG.n>

Receive Equipment Fault/Status Unsolicited Message

For newly Disabled Equipment,
reset Disable Fault Msg after all Packets 0-7 have been sent to reset any open DB Events

18

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Local Copy of Equip Array Tag for LNi, EQi Equipment Disable Msg PKTi

ADD
Source A Equip.CTL.DisablePKTi 0
Source B 1
Dest Equip.CTL.DisablePKTi 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Disable Msg PKTi

GT
Source A Equip.CTL.DisablePKTi 0
Source B 7

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
U

Exit Routine

19

lbl_Exit
LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 231
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault01.WritePtrErr <EDC.RB_Fault01.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest RB_Fault01.WritePtrErrTag <EDC.RB_Fault01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest RB_Fault01.WritePtrErrRec <EDC.RB_Fault01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

Load Equipment Fault/Status Message into Ring Buffer

3

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Fault01.FullBlockWrite <EDC.RB_Fault01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault01.WriteBlockErr <EDC.RB_Fault01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault01.WriteBlockErrCnt <EDC.RB_Fault01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault01.WriteBlockErrCnt <EDC.RB_Fault01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec01[RB_Fault01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec02[RB_Fault01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec03[RB_Fault01.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 1
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault01.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240
Source B 1
Dest RB_Fault01.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 232
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault02.WritePtrErr <EDC.RB_Fault02.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest RB_Fault02.WritePtrErrTag <EDC.RB_Fault02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest RB_Fault02.WritePtrErrRec <EDC.RB_Fault02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Fault02.FullBlockWrite <EDC.RB_Fault02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault02.WriteBlockErr <EDC.RB_Fault02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault02.WriteBlockErrCnt <EDC.RB_Fault02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault02.WriteBlockErrCnt <EDC.RB_Fault02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A: RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr>: 1
Source B: 1

Equipment Data Fault/Status Message

COP
Source: FaultMsg
Dest: EDC_RB_Fault02_Rec01[RB_Fault02.WriteRecPtr]
Length: 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A: RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr>: 1
Source B: 2

Equipment Data Fault/Status Message

COP
Source: FaultMsg
Dest: EDC_RB_Fault02_Rec02[RB_Fault02.WriteRecPtr]
Length: 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A: RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr>: 1
Source B: 3

Equipment Data Fault/Status Message

COP
Source: FaultMsg
Dest: EDC_RB_Fault02_Rec03[RB_Fault02.WriteRecPtr]
Length: 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr>: 1
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr>: 318
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr>: 1
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr>: 317
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2011
Input Par: Equip.CTL.FaultMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 1
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault02.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334
Source B 1
Dest RB_Fault02.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 233
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault03.WritePtrErr <EDC.RB_Fault03.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest RB_Fault03.WritePtrErrTag <EDC.RB_Fault03.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest RB_Fault03.WritePtrErrRec <EDC.RB_Fault03.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault03.FullBlockWrite <EDC.RB_Fault03.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault03.WriteBlockErr <EDC.RB_Fault03.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault03.WriteBlockErrCnt <EDC.RB_Fault03.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault03.WriteBlockErrCnt <EDC.RB_Fault03.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec01[RB_Fault03.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 2

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec02[RB_Fault03.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 3

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec03[RB_Fault03.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteRecPtr
<EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadRecPtr
<EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 1
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault03.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189
Source B 1
Dest RB_Fault03.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 234
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault04.WritePtrErr <EDC.RB_Fault04.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest RB_Fault04.WritePtrErrTag <EDC.RB_Fault04.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest RB_Fault04.WritePtrErrRec <EDC.RB_Fault04.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault04.FullBlockWrite <EDC.RB_Fault04.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault04.WriteBlockErr <EDC.RB_Fault04.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault04.WriteBlockErrCnt <EDC.RB_Fault04.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault04.WriteBlockErrCnt <EDC.RB_Fault04.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec01[RB_Fault04.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec02[RB_Fault04.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec03[RB_Fault04.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.WriteRecPtr
<EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.ReadRecPtr
<EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 1
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault04.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575
Source B 1
Dest RB_Fault04.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 235
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault05.WritePtrErr <EDC.RB_Fault05.WritePtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest RB_Fault05.WritePtrErrTag <EDC.RB_Fault05.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest RB_Fault05.WritePtrErrRec <EDC.RB_Fault05.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault05.FullBlockWrite <EDC.RB_Fault05.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault05.WriteBlockErr <EDC.RB_Fault05.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault05.WriteBlockErrCnt <EDC.RB_Fault05.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault05.WriteBlockErrCnt <EDC.RB_Fault05.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec01[RB_Fault05.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec02[RB_Fault05.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec03[RB_Fault05.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.WriteRecPtr
<EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.ReadRecPtr
<EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 1
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault05.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824
Source B 1
Dest RB_Fault05.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 236
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault06.WritePtrErr <EDC.RB_Fault06.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest RB_Fault06.WritePtrErrTag <EDC.RB_Fault06.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest RB_Fault06.WritePtrErrRec <EDC.RB_Fault06.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault06.FullBlockWrite <EDC.RB_Fault06.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault06.WriteBlockErr <EDC.RB_Fault06.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault06.WriteBlockErrCnt <EDC.RB_Fault06.WriteBlockErrCnt> 8543
Source B 1
Dest RB_Fault06.WriteBlockErrCnt <EDC.RB_Fault06.WriteBlockErrCnt> 8543

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec01[RB_Fault06.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec02[RB_Fault06.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec03[RB_Fault06.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault06.WriteRecPtr
<EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault06.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault06.ReadRecPtr
<EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 1
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault06.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695
Source B 1
Dest RB_Fault06.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Receive Equipment Quality/Health Unsolicited Message

Receive Equipment Quality/Health Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 240
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Quality/Health Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
Block Transaction TImer if Quality Message not enabled -- PdM Sample uses the Quality RB but doesn't use the EqDriver Quality Message

2

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.QualityMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.QualityMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.QualityMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER
EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2104
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A QualityTimeoutER
39960
Source B 1
Dest QualityTimeoutER
39960

MOVE
Source LNi
27
Dest xxQualityLNi
10

MOVE
Source EQi
26
Dest xxQualityEQi
15

Equipment Data Quality/Health Message

FLL
Source 0
Dest QualityMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

PdM - Bypass Quality Tool RxBuffer processing when PdM Sample is ready and valid Tool Config/Msg generated
EqDriver and PdmDriver are sharing the Quality Tool RingBuffer interface

Nov 19, 2022 - When EQ has multple tools with mix of Sample and Trigger, a Triggered Tool can incorrectly use another Tool's Sample signal.
Triggered Tools should not scan the Sample routine. (Moved JSR 249 here and then jumping over RxBuffer/Trigger)

3

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver	Pdm Send Data Collection Sample	Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver	
Equip.CTL.QualityMsg.PdmSampleSend	JSR Routine Name _249_EquipQualityRxPdmSampleSend	Equip.CTL.QualityMsg.PdmSampleSend	lbl_LoadMsg JMP

Receive Equipment Quality/Health Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

v1.6 - Quality Msg Version determined by Header.PacketType 0=v1.5 (5 Data), 1=v1.6 (10 Data)

4

Equipment Data Quality/Health Message Header

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsg.Header
Length 1

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

NE
Source A QualityMsg.Header.PacketType
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
L

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

EQ
Source A QualityMsg.Header.PacketType
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
U

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
/

Equipment Data Quality/Health Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsgv1_5
Length 1

Receive Equipment Quality/Health Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

v1.6 - Quality Msg Version determined by Header.PacketSizer 0=v1.5 (5 Data), 1=v1.6 (10 Data)

5

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsgv1_5.Header.TransNum
7709
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsgv1_5.Footer.TransNum
0
Source B 0

v1.6 - Map v1.5 Quality Message to Message Structure for compatibility

6

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Map v1.5 Quality Message to v1.6 data structure

JSR
Routine Name _246_EquipQualityRxMapV1_5

Receive Equipment Quality/Health Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.QualityMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.QualityMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.EN

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2100
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Receive Equipment Quality/Health Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

8

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.QualityMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.QualityMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Quality/Health Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

10

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER
EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2105
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Diagnostic

ADD
Source A QualityPartialMsgER
125
Source B 1
Dest QualityPartialMsgER
125

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

11

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum
0
Source B 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B QualityMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.QualityMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER

EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2106
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Diagnostic

ADD
Source A QualityTransNumER
4
Source B 1
Dest QualityTransNumER
4

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

12

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum
0
Source B 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum
0
Source B QualityMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.QualityMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Quality Rx DN

EquipStatus01.Update.QualityRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2101
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Receive Equipment Quality/Health Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 1006
Expression	LNi*100+EQi

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	QualityMsg.Header.EQID 0
Source B	ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Equipment ID Mismatch ER
EquipStatus01.Update.QualityRxEqIdER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityMsg.Header.EQID 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.QualityMsg
PerfTrans	EquipStatus03.QualityRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2107
Input Par	Equip.CTL.QualityMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	QualityMsg.Header.TransNum

Diagnostic

ADD	
Source A	QualityEqidER 2112
Source B	1
Dest	QualityEqidER 2112

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A: QualityMsg.Header.TransNum 0
Source B: Equip.CTL.QualityMsg.TransNumLast 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Quality Rx DN
EquipStatus01.Update.QualityRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.QualityMsg.TransNumLast 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.QualityMsg
PerfTrans: EquipStatus03.QualityRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2102
Input Par: Equip.CTL.QualityMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: QualityMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE	
Source A	QualityMsg.Header.TransNum
	0
Source B	Equip.CTL.QualityMsg.TransNumLast
	0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE	
Source A	QualityMsg.Header.TransNum
	0
Source B	Equip.CTL.QualityMsg.TransNumNext
	0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Transaction Number Sequence ER
EquipStatus01.Update.QualityRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.QualityMsg.TransNumNext
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.QualityMsg
PerfTrans	EquipStatus03.QualityRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2109
Input Par	Equip.CTL.QualityMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	QualityMsg.Header.TransNum

Diagnostic

ADD	
Source A	QualitySeqER
	267938
Source B	1
Dest	QualitySeqER
	267938

Diagnostic

MOVE	
Source	LNi
	27
Dest	QualitySeqER_LNi
	9

Diagnostic

MOVE	
Source	EQi
	26
Dest	QualitySeqER_EQi
	36

Diagnostic

MOVE	
Source	QualityMsg.Header.TransNum
	0
Dest	QualitySeqER_TransNum
	7168

Diagnostic

MOVE	
Source	Equip.CTL.QualityMsg.TransNumNext
	0
Dest	QualitySeqER_TransNumNext
	8597

Receive Equipment Quality/Health Unsolicited Message

16

Prepare for next message

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

MOVE
Source QualityMsg.Header.TransNum 0
Dest Equip.CTL.QualityMsg.TransNumLast 0

ADD
Source A QualityMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.QualityMsg.TransNumNext 0

LT
Source A Equip.CTL.QualityMsg.TransNumNext 0
Source B 1

GT
Source A Equip.CTL.QualityMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

MOVE
Source 1
Dest Equip.CTL.QualityMsg.TransNumNext 0

Receive Equipment Quality/Health Unsolicited Message

BTD signed-SINT ToolNum to DINT to support Tools 0-255
used in PdmTrigger Start/Send and Carrier Track

17

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

ToolNum converted from signed-SINT to unsigned

CLR
Dest QualityMsg_Data_ToolNum 26

ToolNum converted from signed-SINT to unsigned

BTD
Source QualityMsg.Data.ToolNum 0
Source Bit 0
Dest QualityMsg_Data_ToolNum 26
Dest Bit 0
Length 8

v2.00 Pdm - Start Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID
Command -13 = PdM Start & Stop/Send Trigger for thies EQID-ToolID (send current data value)

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Command Number

EQ
Source A QualityMsg.Header.CmdNum 0
Source B -11

Equipment Data Quality/Health Message Header - Command Number

EQ
Source A QualityMsg.Header.CmdNum 0
Source B -13

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart
(L)

Pdm Start Data Collection Trigger

JSR
Routine Name _247_EquipQualityRxPdmTriggerStart

Receive Equipment Quality/Health Unsolicited Message

v2.00 Pdm - Send Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID
Command -13 = PdM Start & Stop/Send Trigger for thies EQID-ToolID (send current data value)

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Command Number
EQ
Source A QualityMsg.Header.CmdNum 0
Source B -12

Equipment Data Quality/Health Message Header - Command Number
EQ
Source A QualityMsg.Header.CmdNum 0
Source B -13

Don't send Trigger Commands to RingBuffer
Equipment Data Quality/Health Message Header - Command Number
CLR
Dest QualityMsg.Header.CmdNum 0

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend
L

Pdm Send Data Collection Trigger
JSR
Routine Name _248_EquipQualityRxPdmTriggerSend

PdM - Bypass Quality Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Quality Tool RingBuffer interface

20

lbl_LoadMsg
LBL

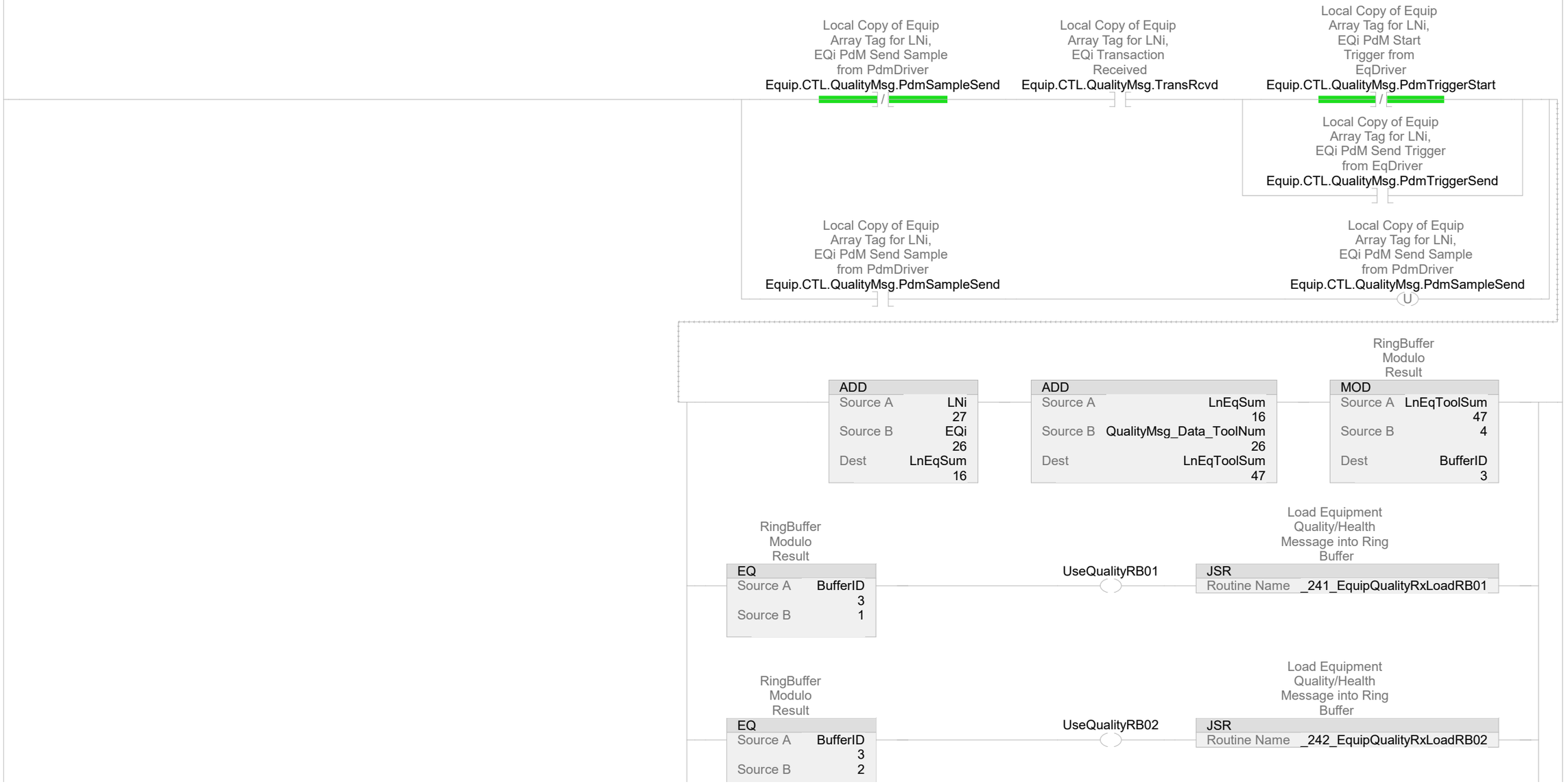
NOP

Receive Equipment Quality/Health Unsolicited Message

Load new message in Ring Buffer
PdM - Ignore PdM Start Trigger Messages (unless Send Trigger also on)
PdM - service PdM Sample Send

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

21



Receive Equipment Quality/Health Unsolicited Message

RingBuffer Modulo Result

EQ
Source A BufferID
Source B 3

UseQualityRB03

Load Equipment Quality/Health Message into Ring Buffer

JSR
Routine Name _243_EquipQualityRxLoadRB03

UseQualityRB01 UseQualityRB02 UseQualityRB03 UseQualityRB04

Load Equipment Quality/Health Message into Ring Buffer

JSR
Routine Name _244_EquipQualityRxLoadRB04

v1.19 -- Load new message in Tracking Conveyor-Carrier arrays
if Tool is configured for Carrier Data
PdM - Ignore Pdm Sample Send Commands and Start Trigger Messages (unless Send Trigger also on)

22

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled for Carrier Tracking Data (from Server Config)
Equip.CFG.QualityMsgTrackEN

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

JSR
Routine Name _245_EquipQualityRxCarrierTrack

Receive Equipment Quality/Health Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

23

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart
U

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.QualityMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.QualityMsg.TransLast
0

Equipment Data Lines

COP
Source QualityMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Quality
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Length CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>

Exit Routine

24

lbl_Exit
LBL

NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 241
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality01.WritePtrErr <EDC.RB_Quality01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest RB_Quality01.WritePtrErrTag <EDC.RB_Quality01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest RB_Quality01.WritePtrErrRec <EDC.RB_Quality01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality01.FullBlockWrite <EDC.RB_Quality01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality01.WriteBlockErr <EDC.RB_Quality01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality01.WriteBlockErrCnt <EDC.RB_Quality01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality01.WriteBlockErrCnt <EDC.RB_Quality01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality01_Rec01[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality01_Rec02[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality01_Rec03[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality01_Rec04[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality01_Rec05[RB_Quality01.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteTagPtr
<EDC.RB_Quality01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteRecPtr
<EDC.RB_Quality01.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadRecPtr
<EDC.RB_Quality01.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 1
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality01.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102
Source B 1
Dest RB_Quality01.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 242
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality02.WritePtrErr <EDC.RB_Quality02.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest RB_Quality02.WritePtrErrTag <EDC.RB_Quality02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest RB_Quality02.WritePtrErrRec <EDC.RB_Quality02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality02.FullBlockWrite <EDC.RB_Quality02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality02.WriteBlockErr <EDC.RB_Quality02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality02.WriteBlockErrCnt <EDC.RB_Quality02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality02.WriteBlockErrCnt <EDC.RB_Quality02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality02_Rec01[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality02_Rec02[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality02_Rec03[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality02_Rec04[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality02_Rec05[RB_Quality02.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr>
984
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr>
983
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 1
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality02.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645
Source B 1
Dest RB_Quality02.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 243
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality03.WritePtrErr <EDC.RB_Quality03.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest RB_Quality03.WritePtrErrTag <EDC.RB_Quality03.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest RB_Quality03.WritePtrErrRec <EDC.RB_Quality03.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality03.FullBlockWrite <EDC.RB_Quality03.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality03.WriteBlockErr <EDC.RB_Quality03.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality03.WriteBlockErrCnt <EDC.RB_Quality03.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality03.WriteBlockErrCnt <EDC.RB_Quality03.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality03_Rec01[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality03_Rec02[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality03_Rec03[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality03_Rec04[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality03_Rec05[RB_Quality03.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr>
1183
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr>
1182
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 1
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Quality03.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565
Source B 1
Dest RB_Quality03.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 244
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality04.WritePtrErr <EDC.RB_Quality04.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest RB_Quality04.WritePtrErrTag <EDC.RB_Quality04.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest RB_Quality04.WritePtrErrRec <EDC.RB_Quality04.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality04.FullBlockWrite <EDC.RB_Quality04.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality04.WriteBlockErr <EDC.RB_Quality04.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality04.WriteBlockErrCnt <EDC.RB_Quality04.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality04.WriteBlockErrCnt <EDC.RB_Quality04.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality04_Rec01[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality04_Rec02[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality04_Rec03[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality04_Rec04[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality04_Rec05[RB_Quality04.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr
<EDC.RB_Quality04.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr
<EDC.RB_Quality04.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 1
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality04.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178
Source B 1
Dest RB_Quality04.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

v1.19 - Store Quality Message Carrier data to Tracking Conveyor-Carrier arrays

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 245
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Scanner Config for matching EQID-ToolID
(combined field allows a quick search)

2

QualityTool Scanner Search (EQID-ToolID)

CPT
Dest ScanSrch 11324
Expression (QualityMsg.Header.EQID *100) + QualityMsg_Data_ToolNum

QualityTool Scanner Search (EQID-ToolID)

NE
Source A ScanSrch 11324
Source B 0

ScanSrchCTRL
RES

MOVE
Source EDC_Track.Cfg.SCN.n 300
Dest ScanSrchCTRL.LEN 300

MOVE
Source 0
Dest ScanSrchCTRL.POS 299

FSC
Control ScanSrchCTRL
Length 300
Position 299
Mode ALL
Expression ScanSrch = EDC_Track.ScanCfg[ScanSrchCTRL.POS].Srch

EN
DN
ER

Exit Routine if Tracking Scanner not Found

3

ScanSrchCTRL.FD
/

lbl_Exit
JMP

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Copy Scanner Configuration to local tags

4

EDC Tracking Scanner Configuration

COP
Source EDC_Track.ScanCfg[ScanSrchCTRL.POS]
Dest ScanCfg
Length 1

Tracking Conveyor Index

MOVE
Source ScanCfg.ConvID
25
Dest CNVi
25

Tracking Equipment Tool Data Index

MOVE
Source ScanCfg.DataID
1
Dest DATi
1

CPT
Dest EQIDi
0
Expression ScanCfg.EQID - (ScanCfg.ConvID * 100)

Exit Routine if Tracking Scanner not Enabled

5

EDC Tracking Scanner Configuration Enabled
ScanCfg.Enable
/

lbl_Exit
JMP

Exit Routine if Invalid Tracking Conveyor Array Index

6

Tracking Conveyor Index

LT
Source A CNVi
25
Source B 0

Tracking Conveyor Index

GT
Source A CNVi
25
Source B EDC_Track.Cfg.CNV.mi
28

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error Count

ADD
Source A EDC_Track.ConvIdERn
0
Source B 1
Dest EDC_Track.ConvIdERn
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error EQID-ToolID

MOVE
Source ScanCfg.Srch
11201
Dest EDC_Track.ConvIdERsrch
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error Index

MOVE
Source CNVi
25
Dest EDC_Track.ConvIdERi
0

lbl_Exit
JMP

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Exit Routine if Invalid Quality Message Tool Data Index
(Configuration indicates which Data word contains the Carrier Number)

7

Tracking Equipment Tool Data Index

LT
Source A DATi 1
Source B 1

Tracking Equipment Tool Data Index

GT
Source A DATi 1
Source B 10

Carrier Tracking from EQ Quality Tools Quality Message Data Error Count

ADD
Source A EDC_Track.DataIdERn 0
Source B 1
Dest EDC_Track.DataIdERn 0

Carrier Tracking from EQ Quality Tools Quality Message Data Error EQID-ToolID

MOVE
Source ScanCfg.Srch 11201
Dest EDC_Track.DataIdERsrch 0

Carrier Tracking from EQ Quality Tools Quality Message Data Error Index

MOVE
Source DATi 1
Dest EDC_Track.DataIdERi 0

lbl_Exit
JMP

Carrier Tracking from EQ Quality Tools Tracking Conveyor

ADD
Source A EDC_Track.Conv[CNVi].DataIdERn 0
Source B 1
Dest EDC_Track.Conv[CNVi].DataIdERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source ScanCfg.Srch 11201
Dest EDC_Track.Conv[CNVi].DataIdERsrch 0

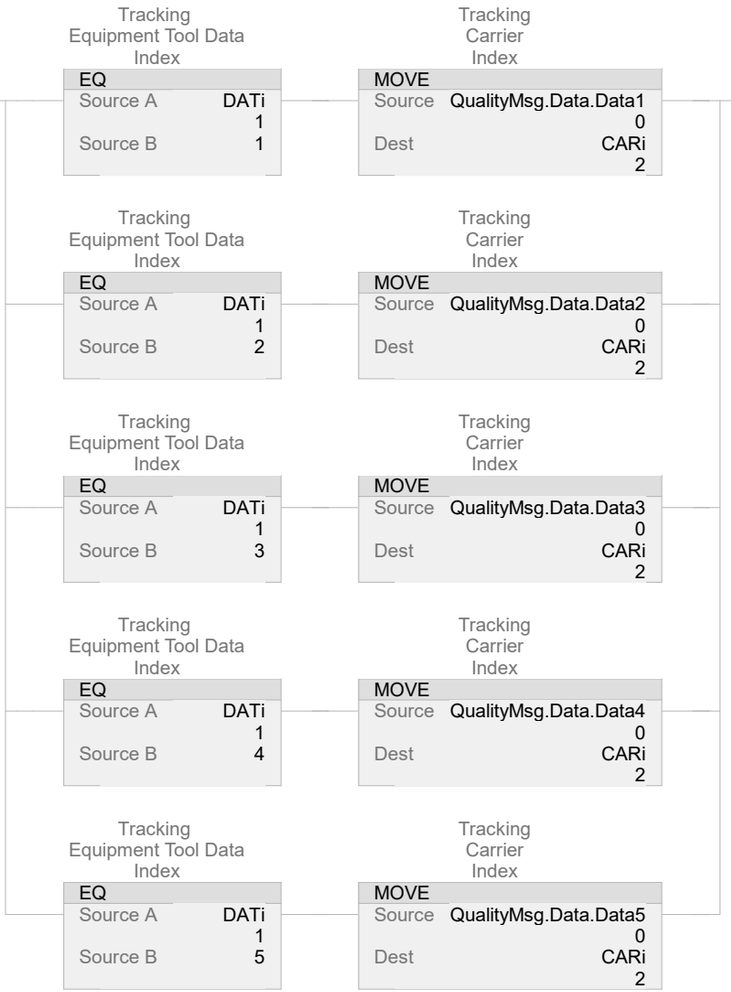
Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source DATi 1
Dest EDC_Track.Conv[CNVi].DataIdERi 0

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Get Carrier Number from Quality Data 1-5

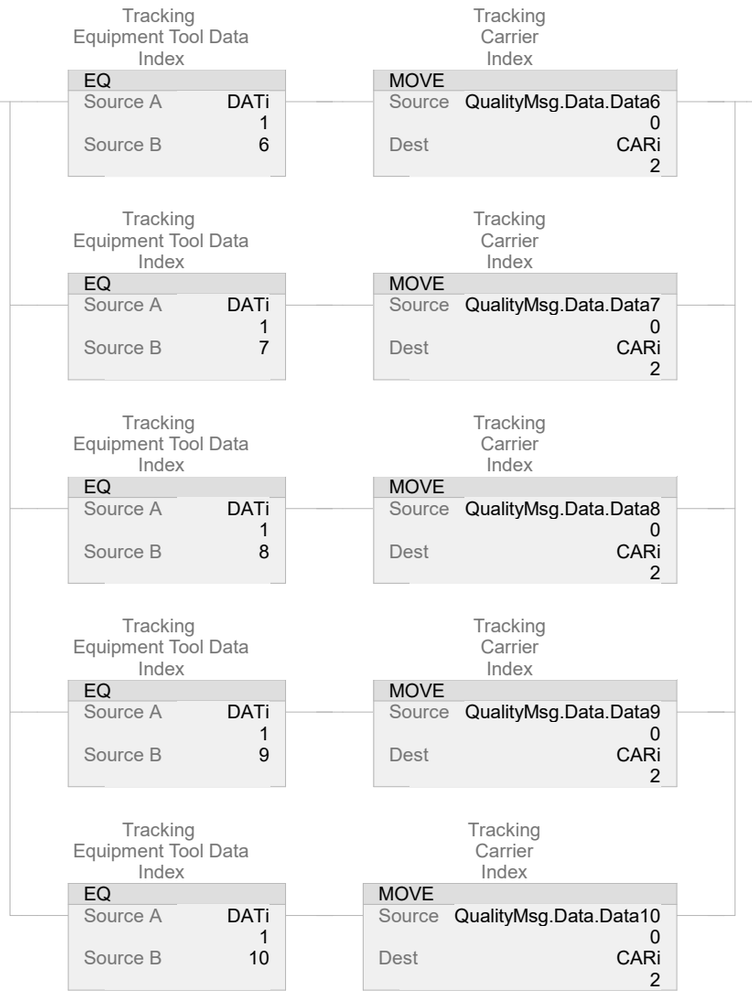
8



Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Get Carrier Number from Quality Data 6-10

9



Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

** Temporary Code to fix Rear Carrier Number **

10

xxFixRearCarrNumEN

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

EQ
Source A QualityMsg.Header.EQID[0]
Source B 113

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

EQ
Source A QualityMsg.Header.EQID[0]
Source B 114

Tracking Carrier Index

LIMIT
Low Limit 1
Test CARi[2]
High Limit 300

Tracking Carrier Index

ADD
Source A CARi[2]
Source B 300
Dest CARi[2]

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Check if valid Carrier Index

11

Tracking Carrier Index

LIMIT	
Low Limit	0
Test	CARi 2
High Limit	EDC_Track.Cfg.CAR.mi 999

ValidCARi

ValidCARi

Tracking Carrier Index

CLR	
Dest	CARi 2

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error Count

ADD	
Source A	EDC_Track.CarrERn 0
Source B	1
Dest	EDC_Track.CarrERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error EQID-ToolID

MOVE	
Source	ScanCfg.Srch 11201
Dest	EDC_Track.CarrERsrch 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error Index

MOVE	
Source	CARi 2
Dest	EDC_Track.CarrERi 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

ADD	
Source A	EDC_Track.Conv[CNVi].CarrERn 0
Source B	1
Dest	EDC_Track.Conv[CNVi].CarrERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE	
Source	ScanCfg.Srch 11201
Dest	EDC_Track.Conv[CNVi].CarrERsrch 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE	
Source	CARi 2
Dest	EDC_Track.Conv[CNVi].CarrERi 0

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Check if valid Equipment Index

12

LIMIT
Low Limit 0
Test EQIDi 0
High Limit CFG.EQ.mi <EDC.CFG.EQ.mi> 99

ValidEQIDi

ValidEQIDi
CLR
Dest EQIDi 0

Diagnostic
ADD
Source A ConvCarr_InvalidEQiCnt 189098
Source B 1
Dest ConvCarr_InvalidEQiCnt 189098

Check if valid Pitch Index

13

ValidEQIDi

MOVE
Source EDC.Line[CNVi].Equip[EQIDi].CFG.Pitch 0
Dest EQPi 0

LIMIT
Low Limit 0
Test EQPi 0
High Limit EDC_Track.Cfg.POS.mi 59

ValidEQPi

ValidEQPi
CLR
Dest EQPi 0

Diagnostic
ADD
Source A ConvCarr_InvalidEQPiCnt 0
Source B 1
Dest ConvCarr_InvalidEQPiCnt 0

Combine Carrier Type (0-4) with Ano & Bno
This allows one Tracking Image to contain multiple Ano/Bno for different Door Types

14

Carrier Type & Ano
for combined search
CPT
Dest TypeAno 230986
Expression (ScanCfg.Type*100000) + QualityMsg.Data.AssyNum

Carrier Type & Bno
for combined search
CPT
Dest TypeBno 275504
Expression (ScanCfg.Type*100000) + QualityMsg.Data.BodyNum

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Search for duplicate Ano
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

15

Equipment Data Quality/Health Message Assembly Sequence Number

NE
Source A QualityMsg.Data.AssyNum
0
Source B 0

Duplicate Ano Search CONTROL
FindAno
RES

Duplicate Ano Search CONTROL

MOVE
Source EDC_Track.Cfg.CAR.n
1000
Dest FindAno.LEN
1000

Duplicate Ano Search CONTROL

FSC
Control FindAno
Length 1000
Position 999
Mode ALL
Expression TypeAno=(EDC_Track.Conv[CNVi].Carr[FindAno.POS].Type*100000+EDC_Track.Conv[CNVi].Carr[FindAno.POS].Ano)
EN
DN
ER

Duplicate Ano Search CONTROL
FindAno.FD

Carrier Tracking from EQ Quality Tools Tracking Conveyor

FLL
Source 0
Dest EDC_Track.Conv[CNVi].Carr[FindAno.POS]
Length 1

Carrier Tracking from EQ Quality Tools Duplicate Ano Deleted Count

ADD
Source A EDC_Track.DupAnoCnt
185
Source B 1
Dest EDC_Track.DupAnoCnt
185

Carrier Tracking from EQ Quality Tools Duplicate Ano Deleted EQID-ToolID

MOVE
Source QualityMsg.Header.EQID
0
Dest EDC_Track.DupAnoSrch
114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Search for duplicate Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

16

Equipment Data Quality/Health Message Assembly Body Number

NE
Source A QualityMsg.Data.BodyNum
0
Source B 0

Duplicate Bno Search CONTROL FindBno
RES

Duplicate Bno Search CONTROL

MOVE
Source EDC_Track.Cfg.CAR.n
1000
Dest FindBno.LEN
1000

Duplicate Bno Search CONTROL

FSC
Control FindBno
Length 1000
Position 999
Mode ALL
Expression TypeBno=(EDC_Track.Conv[CNVi].Carr[FindBno.POS].Type*100000+EDC_Track.Conv[CNVi].Carr[FindBno.POS].Bno)
EN
DN
ER

Duplicate Bno Search CONTROL FindBno.FD

Carrier Tracking from EQ Quality Tools Tracking Conveyor

FLL
Source 0
Dest EDC_Track.Conv[CNVi].Carr[FindBno.POS]
Length 1

Carrier Tracking from EQ Quality Tools Duplicate Bno Deleted Count

ADD
Source A EDC_Track.DupBnoCnt
3
Source B 1
Dest EDC_Track.DupBnoCnt
3

Carrier Tracking from EQ Quality Tools Duplicate Bno Deleted EQID-ToolID

MOVE
Source QualityMsg.Header.EQID
0
Dest EDC_Track.DupBnoSrch
114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Update HMI animation for Model Type
(this provides initial animation for the Door Carriers until they reach the PDC queue - other carriers are typically in PDC or Line tracking before they are scanned)
(0=White, 1=Yellow, 3=Cyan)

17

Shop-specific code
for animation

JSR	
Routine Name	_245_EquipQualityRxCarrierTrack_CfgAnim

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Tracking Conveyor-Carrier Data

18

ValidCARi

Equipment Data Quality/Health Message Program Number Executed

NE
Source A QualityMsg.Data.ActPrgNum
0
Source B 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.ActPrgNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Model
73

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AssyNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Ano
30737

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.BodyNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Bno
75212

Equipment Data Quality/Health Message Program Number Executed

EQ
Source A QualityMsg.Data.ActPrgNum
0
Source B 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AviPrgNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Model
73

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source QualityMsg.TransTime
Dest EDC_Track.Conv[CNVi].Carr[CARi].TransTime
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.CycleCount
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].CycleCount
588517

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source CNVi
25
Dest EDC_Track.Conv[CNVi].Carr[CARi].Loc
12

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source EQPi
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Pos
57

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source Animation
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Anim
4

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source ScanCfg.Type
2
Dest EDC_Track.Conv[CNVi].Carr[CARi].Type
2

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Previous Message Diagnostic

19

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source EDC_Track.Conv[CNVi].CarrCurr
Dest EDC_Track.Conv[CNVi].CarrPrev
Length 1

Store Current Message Diagnostic

20

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source CARi
2
Dest EDC_Track.Conv[CNVi].CarrCurr.Carrier
159

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AssyNum
0
Dest EDC_Track.Conv[CNVi].CarrCurr.Ano
30983

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.BodyNum
0
Dest EDC_Track.Conv[CNVi].CarrCurr.Bno
75438

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source QualityMsg.TransTime
Dest EDC_Track.Conv[CNVi].CarrCurr.TransTime
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.CycleCount
0
Dest EDC_Track.Conv[CNVi].CarrCurr.CycleCount
588763

New Data Count

21

Diagnostic

ADD
Source A ConvCarr_NewDataCnt
4289327
Source B 1
Dest ConvCarr_NewDataCnt
4289327

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Copy EDC_Track (Conv-Carrier) to EDC_TrackAVI (Conv-Position) for AVI Overview Display

22

NOP

Remove duplicate Tracking Conveyor-Position Data for Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

23

Gateway Config
Download Tracking
Allow Duplicate
Carriers on Line
Enable
EDC_CfgGateway.TrackingDupEnable

Equipment Data
Quality/Health
Message Assembly
Sequence Number

NE
Source A QualityMsg.Data.AssyNum
0
Source B 0

FindAnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n
60
Dest FindAnoPitch.LEN
60

FSC
Control FindAnoPitch
Length 60
Position 59
Mode ALL
Expression QualityMsg.Data.AssyNum=EDC_TrackAVI.Line[CNVi].Pitch[FindAnoPitch.POS].Ano
EN
DN
ER

FindAnoPitch.FD

AVI Line Tracking
Line Queue

FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindAnoPitch.POS]
Length 1

AVI Line Tracking
Line Queue

CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindAnoPitch.POS]
4

Diagnostic

ADD
Source A DupAnoPitchCnt
43
Source B 1
Dest DupAnoPitchCnt
43

Diagnostic

MOVE
Source QualityMsg.Header.EQID
0
Dest DupAnoPitchSrch
830

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Remove duplicate Tracking Conveyor-Position Data for Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

24

Gateway Config
Download Tracking
Allow Duplicate
Carriers on Line
Enable
EDC_CfgGateway.TrackingDupEnable

Equipment Data
Quality/Health
Message Assembly
Body Number

NE
Source A QualityMsg.Data.BodyNum
0
Source B 0

FindBnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n
60
Dest FindBnoPitch.LEN
60

FSC
Control FindBnoPitch
Length 60
Position 59
Mode ALL
Expression QualityMsg.Data.BodyNum=EDC_TrackAVI.Line[CNVi].Pitch[FindBnoPitch.POS].Bno
EN
DN
ER

FindBnoPitch.FD

AVI Line Tracking
Line Queue

FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindBnoPitch.POS]
Length 1

AVI Line Tracking
Line Queue

CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindBnoPitch.POS]
4

Diagnostic

ADD
Source A DupBnoPitchCnt
13385
Source B 1
Dest DupBnoPitchCnt
13385

Diagnostic

MOVE
Source QualityMsg.Header.EQID
0
Dest DupBnoPitchSrch
213

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Remove duplicate Tracking Conveyor-Position Data for Cno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

25

Gateway Config Download Tracking Allow Duplicate Carriers on Line Enable
EDC_CfgGateway.TrackingDupEnable

ValidCARi

Tracking Carrier Index
NE
Source A CARi 2
Source B 0

FindCnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n 60
Dest FindCnoPitch.LEN 60

FSC
Control FindCnoPitch
Length 60
Position 59
Mode ALL
Expression CARi=EDC_TrackAVI.Line[CNVi].Pitch[FindCnoPitch.POS].Carrier
EN
DN
ER

FindCnoPitch.FD

AVI Line Tracking Line Queue
FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindCnoPitch.POS]
Length 1

AVI Line Tracking Line Queue
CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindCnoPitch.POS] 4

Diagnostic
ADD
Source A DupCnoPitchCnt 415163
Source B 1
Dest DupCnoPitchCnt 415163

Diagnostic
MOVE
Source QualityMsg.Header.EQID 0
Dest DupCnoPitchSrch 114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Tracking Conveyor-Position Data
If AviPrgNum = 0, load ActPrgNum

26

ValidEQPi

AVI Line Tracking Line Queue
MOVE
Source CARi 2
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Carrier 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.AssyNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Ano 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.BodyNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Bno 0

AVI Line Tracking Line Queue
MOVE
Source Animation 0
Dest EDC_TrackAVI.Line[CNVi].Anim[EQPi] 1

Equipment Data Quality/Health Message Program Number Executed
NE
Source A QualityMsg.Data.ActPrgNum 0
Source B 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.ActPrgNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Model 46

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.AviPrgNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Model 46

Exit Routine Cycle Count

27

lbl_Exit
LBL

Diagnostic
ADD
Source A ConvCarr_CycleCnt 18361303
Source B 1
Dest ConvCarr_CycleCnt 18361303

(End)

Shop-specific code for animation

Update HMI animation for Model Type
(this provides initial animation for the Door Carriers until they reach the PDC queue - other carriers are typically in PDC or Line tracking before they are scanned)
(0=White, 1=Yellow, 3=Cyan)

0

Equipment Data Quality/Health Message Assembly Sequence Number
EQ
Source A QualityMsg.Data.AssyNum 0
Source B 0

Equipment Data Quality/Health Message Assembly Body Number
EQ
Source A QualityMsg.Data.BodyNum 0
Source B 0

Empty

Animation Value
MOVE
Source 1
Dest Animation 0

Empty

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 72

AnimateGH

Animation Value
MOVE
Source 3
Dest Animation 0

Equipment Data Quality/Health Message Program Number from SLCP
EQ
Source A QualityMsg.Data.AviPrgNum 0
Source B 72

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 73

AnimateL

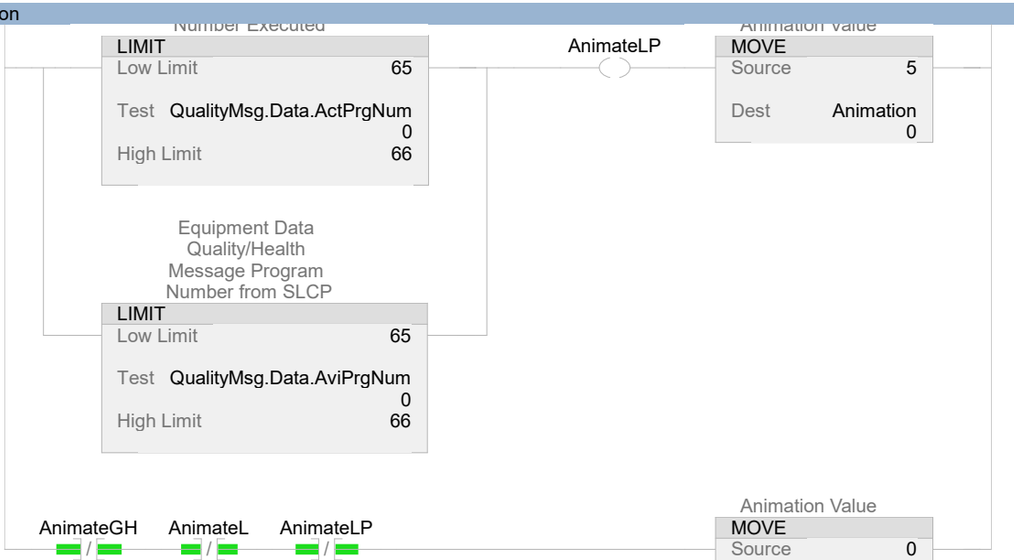
Animation Value
MOVE
Source 4
Dest Animation 0

Equipment Data Quality/Health Message Program Number from SLCP
EQ
Source A QualityMsg.Data.AviPrgNum 0
Source B 73

Equipment Data Quality/Health Message Program Number Executed

Animation Value

Shop-specific code for animation



(End)

Map v1.5 Quality Message to v1.6 data structure

v1.6 - Map v1.5 Quality Message to v1.6 data structure

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	246
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Header, TransTime and Footer

2

Equipment Data Quality/Health Message Header

COP	
Source	QualityMsgv1_5.Header
Dest	QualityMsg.Header
Length	1

Equipment Data Quality/Health Message Transaction Time Stamp

COP	
Source	QualityMsgv1_5.TransTime
Dest	QualityMsg.TransTime
Length	1

Equipment Data Quality/Health Message Footer

COP	
Source	QualityMsgv1_5.Footer
Dest	QualityMsg.Footer
Length	1

Map v1.5 Quality Message to v1.6 data structure

3

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Tool Number

MOVE	
Source	QualityMsgv1_5.Data.ToolNum 38
Dest	QualityMsg.Data.ToolNum 0

Equipment Data Quality/Health Message Program Number from SLCP

MOVE	
Source	QualityMsgv1_5.Data.AviPrgNum 1
Dest	QualityMsg.Data.AviPrgNum 0

Equipment Data Quality/Health Message Program Number Executed

MOVE	
Source	QualityMsgv1_5.Data.ActPrgNum 1
Dest	QualityMsg.Data.ActPrgNum 0

Equipment Data Quality/Health Message Job Status

MOVE	
Source	QualityMsgv1_5.Data.JobStatus 1
Dest	QualityMsg.Data.JobStatus 0

Equipment Data Quality/Health Message Fault Code

MOVE	
Source	QualityMsgv1_5.Data.FaultCode 0
Dest	QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Spare

MOVE	
Source	QualityMsgv1_5.Data.Spare1 2
Dest	QualityMsg.Data.Spare1 0

Equipment Data Quality/Health Message Spare

MOVE	
Source	QualityMsgv1_5.Data.Spare2 0
Dest	QualityMsg.Data.Spare2 0

Equipment Data Quality/Health Message Data Values Used (1-5)

MOVE	
Source	QualityMsgv1_5.Data.DataCount 10
Dest	QualityMsg.Data.DataCount 0

Equipment Data Quality/Health Message Assembly Body Number

MOVE	
Source	QualityMsgv1_5.Data.BodyNum 74759
Dest	QualityMsg.Data.BodyNum 0

Equipment Data Quality/Health Message Assembly Sequence Number

MOVE	
Source	QualityMsgv1_5.Data.AssyNum 29259
Dest	QualityMsg.Data.AssyNum 0

Equipment Data Quality/Health Message VIN Number

COP	
Source	QualityMsgv1_5.Data.VinNum[0]
Dest	QualityMsg.Data.VinNum[0]
Length	20

Equipment Data Quality/Health Message Employee Badge Number for Job Release

COP	
Source	QualityMsgv1_5.Data.BadgeNum[0]
Dest	QualityMsg.Data.BadgeNum[0]
Length	12

Equipment Data Quality/Health Message Cycle Time

MOVE	
Source	QualityMsgv1_5.Data.CycleTime 0
Dest	QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count

MOVE	
Source	QualityMsgv1_5.Data.CycleCount 0
Dest	QualityMsg.Data.CycleCount 0

Map v1.5 Quality Message to v1.6 data structure

4

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 1	Equipment Data Quality/Health Message Quality Data 2	Equipment Data Quality/Health Message Quality Data 3	Equipment Data Quality/Health Message Quality Data 4	Equipment Data Quality/Health Message Quality Data 5
MOVE	MOVE	MOVE	MOVE	MOVE
Source QualityMsgv1_5.Data.Data1 0	Source QualityMsgv1_5.Data.Data2 0	Source QualityMsgv1_5.Data.Data3 0	Source QualityMsgv1_5.Data.Data4 0	Source QualityMsgv1_5.Data.Data5 0
Dest QualityMsg.Data.Data1 0	Dest QualityMsg.Data.Data2 0	Dest QualityMsg.Data.Data3 0	Dest QualityMsg.Data.Data4 0	Dest QualityMsg.Data.Data5 0

5

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 6	Equipment Data Quality/Health Message Quality Data 7	Equipment Data Quality/Health Message Quality Data 8	Equipment Data Quality/Health Message Quality Data 9	Equipment Data Quality/Health Message Quality Data 10
MOVE	MOVE	MOVE	MOVE	MOVE
Source 0	Source 0	Source 0	Source 0	Source 0
Dest QualityMsg.Data.Data6 0	Dest QualityMsg.Data.Data7 0	Dest QualityMsg.Data.Data8 0	Dest QualityMsg.Data.Data9 0	Dest QualityMsg.Data.Data10 0

6

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor	Equipment Data Quality/Health Message Quality Data 2 Scaling Factor	Equipment Data Quality/Health Message Quality Data 3 Scaling Factor	Equipment Data Quality/Health Message Quality Data 4 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE	MOVE	MOVE	MOVE	MOVE
Source QualityMsgv1_5.Data.Scale1 0	Source QualityMsgv1_5.Data.Scale2 0	Source QualityMsgv1_5.Data.Scale3 0	Source QualityMsgv1_5.Data.Scale4 0	Source QualityMsgv1_5.Data.Scale5 0
Dest QualityMsg.Data.Scale1 0	Dest QualityMsg.Data.Scale2 0	Dest QualityMsg.Data.Scale3 0	Dest QualityMsg.Data.Scale4 0	Dest QualityMsg.Data.Scale5 0

Map v1.5 Quality Message to v1.6 data structure

Data
Move individual elements to resolve data differences

7

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE	MOVE	MOVE	MOVE	MOVE
Source 0	Source 0	Source 0	Source 0	Source 0
Dest QualityMsg.Data.Scale6 0	Dest QualityMsg.Data.Scale7 0	Dest QualityMsg.Data.Scale8 0	Dest QualityMsg.Data.Scale9 0	Dest QualityMsg.Data.Scale10 0

(End)

Pdm Start Data Collection Trigger

v2.00 Pdm - Start Trigger
Command -11 = Start Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 247
Dest Prog.Routine225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine]-1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine]-1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolStartTrigSrch944074
Expression (QualityMsg.Header.EQID*1000)+QualityMsg_Data_ToolNum

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A ToolStartTrigSrch944074
Source B 0

Tool Search for Start Trigger ToolStartTrigFSC

RES

Tool Search for Start Trigger

ADD
Source A PdM_Cfg.TL.ei914
Source B 1
Dest ToolStartTrigFSC.LEN915

Tool Search for Start Trigger

MOVE
Source 0
Dest ToolStartTrigFSC.POS496

Tool Search for Start Trigger

FSC
Control ToolStartTrigFSC
Length 915
Position 496
Mode ALL
Expression ToolStartTrigSrch=PdM_Tool[ToolStartTrigFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Start Data Collection Trigger

Exit Routine if Pdm Tool not Found

3

Tool Search for Start Trigger
ToolStartTrigFSC.FD
/

lbl_Exit
JMP

Exit Routine if
Pdm Tool not Enabled
Pdm Tool Trigger not Enabled

(keep this separate from previous rung to avoid bad index when Tool not found)

4

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Cfg.Enable
/

lbl_Exit
JMP

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Cfg.TriggerEnable
/

Set EQID-Tool Start Trigger

5

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Ctl.TriggerStart
L

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Trigger

v2.00 Pdm - Send Trigger
Command -12 = Send Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 248
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolSendTrigSrch 1010021
Expression (QualityMsg.Header.EQID*1000)+QualityMsg_Data_ToolNum

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A ToolSendTrigSrch 1010021
Source B 0

Tool Search for Send Trigger

ToolSendTrigFSC
RES

Tool Search for Send Trigger

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest ToolSendTrigFSC.LEN 915

Tool Search for Send Trigger

MOVE
Source 0
Dest ToolSendTrigFSC.POS 827

Tool Search for Send Trigger

FSC
Control ToolSendTrigFSC
Length 915
Position 827
Mode ALL
Expression ToolSendTrigSrch=PdM_Tool[ToolSendTrigFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Send Data Collection Trigger

Exit Routine if Pdm Tool not Found

3

Tool Search for Send Trigger
ToolSendTrigFSC.FD
/

lbl_Exit
JMP

Exit Routine if
Pdm Tool not Enabled
Pdm Tool Trigger not Enabled

(keep this separate from previous rung to avoid bad index when Tool not found)

4

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.Enable
/

lbl_Exit
JMP

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.TriggerEnable
/

Set EQID-Tool Send Trigger

5

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Ctl.TriggerSend
L

Load Pdm Data 1-10 to Quality Message

6

PdM Data Loop Index
CLR
Dest LOOPi
10

Load Pdm Data 1-10 to Quality Message
Skip to next Data if this Data not Enabled

7

lbl_DataLoop
LBL

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.Data[LOOPi].Enable
/

lbl_NextData
JMP

Pdm Send Data Collection Trigger

Load Data Value

8

MOVE
Source PdM_Tool[ToolSendTrigFSC.POS].Array[LOOPi].Data
??
Dest PdmData
64

MOVE
Source PdM_Tool[ToolSendTrigFSC.POS].Array[LOOPi].Scale
??
Dest PdmScale
1000

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data1
0

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale1
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 1

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data2
0

Equipment Data Quality/Health Message Quality Data 2 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale2
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 2

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data3
0

Equipment Data Quality/Health Message Quality Data 3 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale3
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 3

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data4
0

Equipment Data Quality/Health Message Quality Data 4 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale4
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 4

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale5
0

Pdm Send Data Collection Trigger



Pdm Send Data Collection Trigger

Increment Loop Counter

9

lbl_NextData
LBL

PdM Data Loop Index
ADD
Source A LOOPi 10
Source B 1
Dest LOOPi 10

PdM Data Loop Index
LT
Source A LOOPi 10
Source B 10

lbl_DataLoop
JMP

Exit Routine

10

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Sample

v2.00 Pdm - Load Sample to Quality Message
Pdm Driver sets SampleSend and ToolID for Quality Tool to Send

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 249
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

PdM Tool sets SampleSend and ToolID

2

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolSendSmplSrch 79032
Expression (((LNi*100)+EQi)*1000)+Equip.CTL.QualityMsg.PdmSampleToolID

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

NE
Source A ToolSendSmplSrch 79032
Source B 0

Tool Search for Send Sample

ToolSendSmplFSC
RES

Tool Search for Send Sample

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest ToolSendSmplFSC.LEN 915

Tool Search for Send Sample

MOVE
Source 0
Dest ToolSendSmplFSC.POS 532

Tool Search for Send Sample

FSC
Control ToolSendSmplFSC
Length 915
Position 532
Mode ALL
Expression ToolSendSmplSrch=PdM_Tool[ToolSendSmplFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Send Data Collection Sample

Exit Routine if Pdm Tool not Found

3

Tool Search for Send Sample
ToolSendSmplFSC.FD
/

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend
U

lbl_Exit
JMP

Exit Routine if Pdm Tool not Enabled

4

PdM Tool Array
PdM_Tool[ToolSendSmplFSC.POS].Cfg.Enable
/

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend
U

lbl_Exit
JMP

Initialize QualityMsg
PdM Sample generates its own Header, etc.

5

Equipment Data Quality/Health Message

FLL	
Source	0
Dest	QualityMsg
Length	1

Pdm Send Data Collection Sample

Increment Transaction Number
Pdm uses Transaction Numbers 10,000-10,999 to avoid conflicts with EqDriver

6

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

ADD
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 1
Dest: Equip.CTL.QualityMsg.PdmSampleTransNum 0

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

LT
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 10001

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

GT
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 10999

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

MOVE
Source: 10001
Dest: Equip.CTL.QualityMsg.PdmSampleTransNum 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Dest: QualityMsg.Header.TransNum 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Dest: QualityMsg.Footer.TransNum 0

Transaction Time

7

Equipment Data Quality/Health Message Transaction Time Stamp

COP
Source: CLX.TransTime <EDC.CLX.TransTime>
Dest: QualityMsg.TransTime
Length: 1

Pdm Send Data Collection Sample

Equipment ID & Tool Number

8

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

MOVE
Source 1
Dest QualityMsg.Header.PacketType
0

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Cfg.EQID
79
Dest QualityMsg.Header.EQID
0

Equipment Data Quality/Health Message Tool Number

BTD
Source PdM_Tool[ToolSendSmplFSC.POS].Cfg.ToolID
32
Source Bit 0
Dest QualityMsg.Data.ToolNum
0
Dest Bit 0
Length 8

Job Status

9

Equipment Data Quality/Health Message Job Status

MOVE
Source 1
Dest QualityMsg.Data.JobStatus
0

Load Pdm Data 1-10 to Quality Message

10

PdM Data Loop Index

CLR
Dest LOOPi
10

Load Pdm Data 1-10 to Quality Message
Skip to next Data if this Data not Enabled

11

lbl_DataLoop
LBL

PdM Tool Array
PdM_Tool[ToolSendSmplFSC.POS].Cfg.Data[LOOPi].Enable
/

lbl_NextData
JMP

Pdm Send Data Collection Sample

12

Load Data Value
(default to Average if illegal Type)

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Array[LOOPi].Data
??
Dest PdmData
64

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Array[LOOPi].Scale
??
Dest PdmScale
1000

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data1
0

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale1
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 1
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 1

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data2
0

Equipment Data Quality/Health Message Quality Data 2 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale2
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 2

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data3
0

Equipment Data Quality/Health Message Quality Data 3 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale3
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 3
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 3

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data4
0

Equipment Data Quality/Health Message Quality Data 4 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale4
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount
0

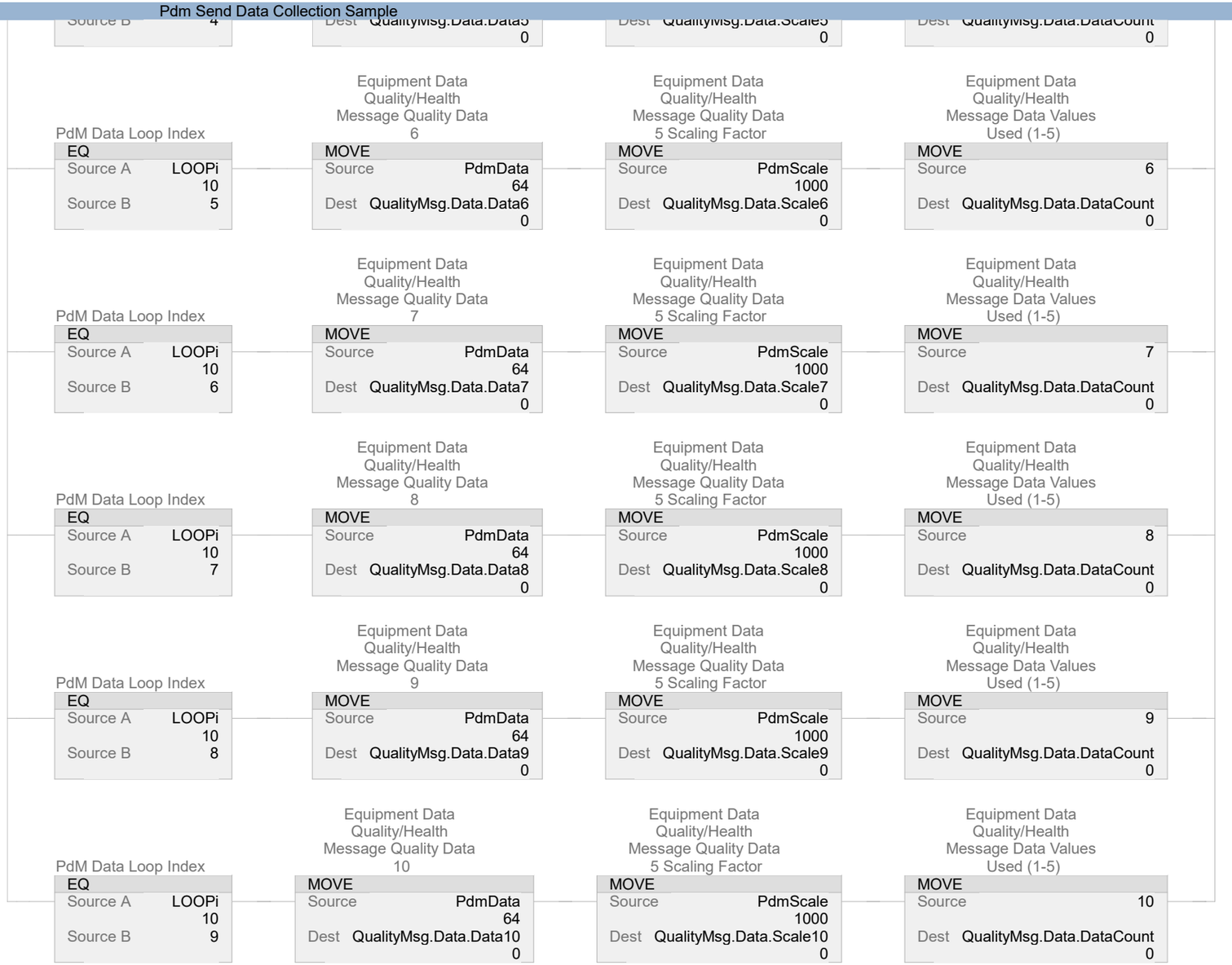
PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 4

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data5

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale5

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 5
Dest QualityMsg.Data.DataCount

Pdm Send Data Collection Sample



Pdm Send Data Collection Sample

Increment Loop Counter

13

lbl_NextData
LBL

PdM Data Loop Index
ADD
Source A LOOPi 10
Source B 1
Dest LOOPi 10

PdM Data Loop Index
LT
Source A LOOPi 10
Source B 10

lbl_DataLoop
JMP

Exit Routine

14

lbl_Exit
LBL

NOP

(End)

Receive Equipment Trend Unsolicited Message

Receive Equipment Trend Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 250
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Trend Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
Block the Transaction TImer if Trend Message not enabled -- PdM Sample uses the Trend RB but doesn't use the EqDriver Trend Message

2

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.TrendMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.TrendMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.TrendMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER
EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2204
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

MOVE
Source LNi
27
Dest xxTrendLNi
0

MOVE
Source EQi
26
Dest xxTrendEQi
53

Equipment Data Trend Message

FLL
Source 0
Dest TrendMsg
Length 1

lbl_MsgReset
JMP

PdM - Bypass Trend Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Trend Tool RingBuffer interface

3

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

lbl_PdmSample
JMP

Receive Equipment Trend Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

4

Equipment Data
Trend Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Dest TrendMsg
Length 1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data
Trend Message
Header - Transaction
Number (must match
Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data
Trend Message
Footer - Transaction
Number (must match
Footer)

EQ
Source A TrendMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Receive Equipment Trend Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.TrendMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.TrendMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source TrendMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.EN

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2200
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Receive Equipment Trend Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE
Source A TrendMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.TrendMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.TrendMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Trend Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER
EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2205
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B 0
0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE
Source A TrendMsg.Footer.TransNum
0
Source B 0
0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B TrendMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.TrendMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER

EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2206
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum
	0
Source B	0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Footer.TransNum
	0
Source B	0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ	
Source A	TrendMsg.Header.TransNum
	0
Source B	TrendMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.TrendMsg.TransRcvd (L)

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Trend Rx DN

EquipStatus01.Update.TrendRxOK (L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.DN

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2201
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

Receive Equipment Trend Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Expected Equipment ID

CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A TrendMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Equipment ID Mismatch ER
EquipStatus01.Update.TrendRxEqIdER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2207
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B Equip.CTL.TrendMsg.TransNumLast
0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Trend Rx DN
EquipStatus01.Update.TrendRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Equip.CTL.TrendMsg.TransNumLast
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.KA

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2202
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum 0
Source B	Equip.CTL.TrendMsg.TransNumLast 0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum 0
Source B	Equip.CTL.TrendMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Transaction Number Sequence ER
EquipStatus01.Update.TrendRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.TrendMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2209
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

Diagnostic

ADD	
Source A	TrendSeqER 43570
Source B	1
Dest	TrendSeqER 43570

Diagnostic

MOVE	
Source	LNi 27
Dest	TrendSeqER_LNi 5

Diagnostic

MOVE	
Source	EQi 26
Dest	TrendSeqER_EQi 25

Diagnostic

MOVE	
Source	TrendMsg.Header.TransNum 0
Dest	TrendSeqER_TransNum 5639

Diagnostic

MOVE	
Source	Equip.CTL.TrendMsg.TransNumNext 0
Dest	TrendSeqER_TransNumNext 20

Receive Equipment Trend Unsolicited Message

v2.00 Pdm - Start Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Spare
EQ
Source A TrendMsg.Data.Spare1[0]
Source B -11

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart
L

Pdm Start Data Collection Trigger
JSR
Routine Name _257_EquipTrendRxPdmTriggerStart

v2.00 Pdm - Send Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID

Don't send Trigger Commands to RingBuffer

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Spare
EQ
Source A TrendMsg.Data.Spare1[0]
Source B -12

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerSend
L

Pdm Send Data Collection Trigger
JSR
Routine Name _258_EquipTrendRxPdmTriggerSend

PdM - Bypass Trend Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Trend Tool RingBuffer interface

17

lbl_PdmSample
LBL

NOP

v2.00 Pdm - Load Sample to Trend Message
Pdm Driver sets SampleSend and ToolID for Trend Tool to Send

18

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Pdm Send Data Collection Sample
JSR
Routine Name _259_EquipTrendRxPdmSampleSend

Receive Equipment Trend Unsolicited Message

Load new message in Ring Buffer
PdM - Ignore PdM Start Trigger Messages
PdM - service PdM Sample Send

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

U

ADD
Source A: LNi 27
Source B: EQi 26
Dest: LnEqSum 16

ADD
Source A: LnEqSum 16
Source B: TrendMsg.Data.ToolNum 0
Dest: LnEqToolSum 47

RingBuffer Modulo Result
MOD
Source A: LnEqToolSum 47
Source B: 2
Dest: BufferID 3

RingBuffer Modulo Result
EQ
Source A: BufferID 3
Source B: 1

UseTrendRB01

Load Equipment Trend Message into Ring Buffer
JSR
Routine Name: _251_EquipTrendRxLoadRB01

UseTrendRB01

UseTrendRB02

Load Equipment Trend Message into Ring Buffer
JSR
Routine Name: _252_EquipTrendRxLoadRB02

Receive Equipment Trend Unsolicited Message

Prepare for next message

20

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source TrendMsg.Header.TransNum
0
Dest Equip.CTL.TrendMsg.TransNumLast
0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A TrendMsg.Header.TransNum
0
Source B 1
Dest Equip.CTL.TrendMsg.TransNumNext
0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.TrendMsg.TransNumNext
0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.TrendMsg.TransNumNext
0
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.TrendMsg.TransNumNext
0

Receive Equipment Trend Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

21

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart
U

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerSend
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.TrendMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.TrendMsg.TransLast
0

Equipment Data Lines

COP
Source TrendMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Trend
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Length CFG.TrendMSG.n
<EDC.CFG.TrendMSG.n>

Exit Routine

22

lbl_Exit
LBL

NOP

(End)

Load Equipment Trend Message into Ring Buffer

Load Equipment Trend Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 251
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Trend Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Trend01.WritePtrErr <EDC.RB_Trend01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest RB_Trend01.WritePtrErrTag <EDC.RB_Trend01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest RB_Trend01.WritePtrErrRec <EDC.RB_Trend01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2213
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

Load Equipment Trend Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Trend01.FullBlockWrite <EDC.RB_Trend01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Trend01.WriteBlockErr <EDC.RB_Trend01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Trend01.WriteBlockErrCnt <EDC.RB_Trend01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Trend01.WriteBlockErrCnt <EDC.RB_Trend01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2214
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Trend Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

	EDC Ring Buffer Control Next Write Tag	Equipment Data Trend Message
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 1	COP Source TrendMsg Dest EDC_RB_Trend01_Rec01[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 2	COP Source TrendMsg Dest EDC_RB_Trend01_Rec02[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 3	COP Source TrendMsg Dest EDC_RB_Trend01_Rec03[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 4	COP Source TrendMsg Dest EDC_RB_Trend01_Rec04[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 5	COP Source TrendMsg Dest EDC_RB_Trend01_Rec05[RB_Trend01.WriteRecPtr] Length 1

Load Equipment Trend Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteTagPtr
<EDC.RB_Trend01.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteRecPtr
<EDC.RB_Trend01.WriteRecPtr>
326
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadRecPtr
<EDC.RB_Trend01.ReadRecPtr>
325
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2211
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Load Equipment Trend Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 1
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

Load Equipment Trend Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Trend01.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413
Source B 1
Dest RB_Trend01.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Trend Message into Ring Buffer

Load Equipment Trend Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 252
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Trend Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Trend02.WritePtrErr <EDC.RB_Trend02.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest RB_Trend02.WritePtrErrTag <EDC.RB_Trend02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest RB_Trend02.WritePtrErrRec <EDC.RB_Trend02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2213
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

Load Equipment Trend Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Trend02.FullBlockWrite <EDC.RB_Trend02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Trend02.WriteBlockErr <EDC.RB_Trend02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Trend02.WriteBlockErrCnt <EDC.RB_Trend02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Trend02.WriteBlockErrCnt <EDC.RB_Trend02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2214
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Trend Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

	EDC Ring Buffer Control Next Write Tag	Equipment Data Trend Message
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 1	COP Source TrendMsg Dest EDC_RB_Trend02_Rec01[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 2	COP Source TrendMsg Dest EDC_RB_Trend02_Rec02[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 3	COP Source TrendMsg Dest EDC_RB_Trend02_Rec03[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 4	COP Source TrendMsg Dest EDC_RB_Trend02_Rec04[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 5	COP Source TrendMsg Dest EDC_RB_Trend02_Rec05[RB_Trend02.WriteRecPtr] Length 1

Load Equipment Trend Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteRecPtr
<EDC.RB_Trend02.WriteRecPtr>
202
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadRecPtr
<EDC.RB_Trend02.ReadRecPtr>
201
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2211
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Load Equipment Trend Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 1
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

Load Equipment Trend Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Trend02.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674
Source B 1
Dest RB_Trend02.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Pdm Start Data Collection Trigger

v2.00 Pdm - Start Trigger
Command -11 = Start Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 257
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendStartTrigSrch 53002
Expression TrendMsg.Header.EQID*1000+TrendMsg.Data.ToolNum

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A TrendStartTrigSrch 53002
Source B 0

Trend Search for Start Trigger TrendStartTrigFSC

RES

Trend Search for Start Trigger

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendStartTrigFSC.LEN 4

Trend Search for Start Trigger

MOVE
Source 0
Dest TrendStartTrigFSC.POS 1

Trend Search for Start Trigger

FSC
Control TrendStartTrigFSC
Length 4
Position 1
Mode ALL
Expression TrendStartTrigSrch=PdM_Trend[TrendStartTrigFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Start Data Collection Trigger

Exit Routine if Pdm Trend not Found

3

Trend Search for Start Trigger
TrendStartTrigFSC.FD
/

lbl_Exit
JMP

Exit Routine if Pdm Trend not Enabled

4

PdM Trend Array
PdM_Trend[TrendStartTrigFSC.POS].Cfg.Enable
/

lbl_Exit
JMP

Set EQID-Tool Start Trigger

5

PdM Trend Array
PdM_Trend[TrendStartTrigFSC.POS].Ctl.TriggerStart
L

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Trigger

v2.00 Pdm - Send Trigger
Command -12 = Send Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 258
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendSendTrigSrch 53002
Expression TrendMsg.Header.EQID*1000+TrendMsg.Data.ToolNum

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A TrendSendTrigSrch 53002
Source B 0

Trend Search for Send Trigger

TrendSendTrigFSC
RES

Trend Search for Send Trigger

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendSendTrigFSC.LEN 4

Trend Search for Send Trigger

MOVE
Source 0
Dest TrendSendTrigFSC.POS 1

Trend Search for Send Trigger

FSC
Control TrendSendTrigFSC
Length 4
Position 1
Mode ALL
Expression TrendSendTrigSrch=PdM_Trend[TrendSendTrigFSC.POS].Cfg.EqToolID
EN
DN
ER

Pdm Send Data Collection Trigger

Exit Routine if Pdm Trend not Found

3

Trend Search for Send Trigger
TrendSendTrigFSC.FD
/
lbl_Exit
JMP

Exit Routine if Pdm Trend not Enabled

4

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Cfg.Enable
/
lbl_Exit
JMP

Exit Routine if Pdm Data not Enabled

5

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Cfg.Data.Enable
/
lbl_Exit
JMP

Set EQID-Tool Send Trigger

6

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Ctl.TriggerSend
L

Load Data Values

7

Equipment Data Trend Message Trend Data

COP	
Source	PdM_Trend[TrendSendTrigFSC.POS].Array.Data[0]
Dest	TrendMsg.Data.Trend[0]
Length	120

Equipment Data Trend Message Trend Scale

MOVE	
Source	PdM_Trend[TrendSendTrigFSC.POS].Array.Scale
	1000
Dest	TrendMsg.Data.Scale
	0

Equipment Data Trend Message Data Values Used (1-120)

MOVE	
Source	PdM_Trend[TrendSendTrigFSC.POS].Ctl.DataCount
	15
Dest	TrendMsg.Data.DataCount
	0

Exit Routine

8

lbl_Exit
LBL
NOP

(End)

Pdm Send Data Collection Sample

v2.00 Pdm - Load Sample to Trend Message
Pdm Driver sets SampleSend and ToolID for Trend Tool to Send

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 259
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

PdM Tool sets SampleSend and ToolID

2

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendSendSmplSrch 100005
Expression (((LNi*100)+EQi)*1000)+Equip.CTL.TrendMsg.PdmSampleToolID

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

NE
Source A TrendSendSmplSrch 100005
Source B 0

Trend Search for Send Sample TrendSendSmplFSC

RES

Trend Search for Send Sample

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendSendSmplFSC.LEN 12

Trend Search for Send Sample

MOVE
Source 0
Dest TrendSendSmplFSC.POS 6

Trend Search for Send Sample

FSC
Control TrendSendSmplFSC
Length 12
Position 6
Mode ALL
Expression TrendSendSmplSrch=PdM_Trend[TrendSendSmplFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Send Data Collection Sample

Exit Routine if Pdm Trend not Found

3

Trend Search for
Send Sample
TrendSendSmplFSC.FD

lbl_Exit
JMP

Exit Routine if Pdm Trend not Enabled

4

PdM Trend Array
PdM_Trend[TrendSendSmplFSC.POS].Cfg.Enable

lbl_Exit
JMP

Exit Routine if Pdm Data not Enabled

5

PdM Trend Array
PdM_Trend[TrendSendSmplFSC.POS].Cfg.Data.Enable

lbl_Exit
JMP

Initialize TrendMsg
PdM Sample generates its own Header, etc.

6

Equipment Data
Trend Message

FLL	
Source	0
Dest	TrendMsg
Length	1

Pdm Send Data Collection Sample

Increment Transaction Number
Pdm uses Transaction Numbers 10,000-10,999 to avoid conflicts with EqDriver

7

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

ADD
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 1
Dest: Equip.CTL.TrendMsg.PdmSampleTransNum 0

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

LT
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 10001

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

GT
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 10999

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

MOVE
Source: 10001
Dest: Equip.CTL.TrendMsg.PdmSampleTransNum 0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Dest: TrendMsg.Header.TransNum 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Dest: TrendMsg.Footer.TransNum 0

Transaction Time

8

Equipment Data Trend Message Transaction Time Stamp

COP
Source: CLX.TransTime <EDC.CLX.TransTime>
Dest: TrendMsg.TransTime
Length: 1

Pdm Send Data Collection Sample

Equipment ID & Tool Number

9

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Cfg.EQID 0
Dest TrendMsg.Header.EQID 0

Equipment Data Trend Message Tool Number

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Cfg.ToolID 0
Dest TrendMsg.Data.ToolNum 0

Job Status

10

Equipment Data Trend Message Job Status

MOVE
Source 1
Dest TrendMsg.Data.JobStatus 0

Load Data Values

Note: Rate is an INTEGER and doesn't support sample rates >1/second
(1 is loaded regardless of the Rate setting)

11

Equipment Data Trend Message Trend Data

COP
Source PdM_Trend[TrendSendSmplFSC.POS].Array.Data[0]
Dest TrendMsg.Data.Trend[0]
Length 120

Equipment Data Trend Message Trend Scale

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Array.Scale 0
Dest TrendMsg.Data.Scale 0

Equipment Data Trend Message Data Values Used (1-120)

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Ctl.DataCount 0
Dest TrendMsg.Data.DataCount 0

Equipment Data Trend Message Trend Rate - samples per second (1-10)

MOVE
Source 1
Dest TrendMsg.Data.Rate 0

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Receive Equipment Version Unsolicited Message

Receive Equipment Version Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 260
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

2

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.VersionMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.VersionMsg.TransTMR
Preset 999999999
Accum 948907
EN
DN

Copy Rx Buffer to local registers
DINT Array to UDT

3

Equipment Data Version Message Format

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Dest VersionMsg
Length 1

Receive Equipment Version Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

4

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

5

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.EN
/

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.VersionMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.VersionMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source VersionMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2300
Input Par Equip.CTL.VersionMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

6

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE
Source A VersionMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.VersionMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.VersionMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Version Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset
L

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2305
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Version Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

9

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum
	0
Source B	0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Footer.TransNum
	0
Source B	0

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum
	0
Source B	VersionMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.VersionMsg.TransReset

(L)

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2306
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset

(JMP)

Receive Equipment Version Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

10

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE
Source A VersionMsg.Header.TransNum
0
Source B 0
0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE
Source A VersionMsg.Footer.TransNum
0
Source B 0
0

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Header.TransNum
0
Source B VersionMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd
L

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source VersionMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2301
Input Par Equip.CTL.VersionMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

11

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 1006
Expression	LNi*100+EQi

Equipment Data Version Message Format Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	VersionMsg.Header.EQID 0
Source B	ExpectedEQID 1006

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Header.EQID 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2307
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ	
Source A	VersionMsg.Header.TransNum 0
Source B	Equip.CTL.VersionMsg.TransNumLast 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.VersionMsg.TransNumLast 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2302
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Version Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE
- Source A: VersionMsg.Header.TransNum 0
- Source B: Equip.CTL.VersionMsg.TransNumNext 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
- Source: Equip.CTL.VersionMsg.TransNumNext 0
- Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR
- Routine Name: _901_TransLogSBR
- Input Par: 2309
- Input Par: Equip.CTL.VersionMsg.TransTMR.ACC
- Input Par: LNi
- Input Par: EQi
- Input Par: VersionMsg.Header.TransNum

Load new message in Version Table

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Equipment Data Lines

COP
- Source: VersionMsg
- Dest: EDC_EqRxData.Line[LNi].Equip[EQi].Version
- Length: 1

Log Transactions

JSR
- Routine Name: _901_TransLogSBR
- Input Par: 2311
- Input Par: Equip.CTL.VersionMsg.TransTMR.ACC
- Input Par: LNi
- Input Par: EQi
- Input Par: VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Prepare for next message

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source VersionMsg.Header.TransNum 0
Dest Equip.CTL.VersionMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A VersionMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.VersionMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.VersionMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.VersionMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.VersionMsg.TransNumNext 0

Receive Equipment Version Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

16

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.VersionMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.VersionMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.VersionMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.VersionMsg.TransLast
0

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Length CFG.VersionMSG.n
<EDC.CFG.VersionMSG.n>

Exit Routine

17

lbl_Exit
LBL

NOP

(End)

Receive Equipment Cycle Unsolicited Message

Receive Equipment Quality/Health Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	270
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Cycle Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

2

Local Copy of Equip Array Tag for LNi, EQi Cycle Message Enabled (from Driver Callout Msg)
Equip.CFG.CycleMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.CycleMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.CycleMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CycleMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2504
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

MOVE
Source LNi
27
Dest CycleLNi
10

Diagnostic

MOVE
Source EQi
26
Dest CycleEQi
14

Equipment Data Cycle Message

FLL
Source 0
Dest CycleMsg
Length 1

lbl_MsgReset
JMP

Copy Rx Buffer to local registers
DINT Array to UDT

3

Equipment Data Cycle Message

CPS
Source EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Dest CycleMsg
Length 1

Receive Equipment Cycle Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

4

Equipment Data
Cycle Message Header
- Transaction Number
(must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data
Cycle Message Footer
- Transaction Number
(must match Footer)

EQ
Source A CycleMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

5

Local Copy of Equip
Array Tag for LNi,
EQi Partial
MessageTimer
Equip.CTL.CycleMsg.PartialTMR.EN

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Timer
Equip.CTL.CycleMsg.TransTMR
RES

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Timer Accum Last
(used to calculate
differential)

CLR
Dest Equip.CTL.CycleMsg.TransLast
0

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CycleMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2500
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Receive Equipment Cycle Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

6

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE
Source A CycleMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.CycleMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.CycleMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Cycle Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2505
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

9

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum
	0
Source B	0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Footer.TransNum
	0
Source B	0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum
	0
Source B	CycleMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.CycleMsg.TransReset (L)

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER

EquipStatus01.Update.CycleRxER (L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CycleMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CycleMsg
PerfTrans	EquipStatus04.CycleRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2506
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

lbl_MsgReset (JMP)

Receive Equipment Cycle Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

10

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE
Source A CycleMsg.Footer.TransNum
0
Source B 0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B CycleMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.CycleMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Cycle Rx DN

EquipStatus01.Update.CycleRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2501
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Receive Equipment Cycle Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

11

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Expected Equipment ID

CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Cycle Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A CycleMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2507
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A: CycleMsg.Header.TransNum 0
Source B: Equip.CTL.CycleMsg.TransNumLast 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Cycle Rx DN
EquipStatus01.Update.CycleRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.CycleMsg.TransNumLast 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.CycleMsg
PerfTrans: EquipStatus04.CycleRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2502
Input Par: Equip.CTL.CycleMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum 0
Source B	Equip.CTL.CycleMsg.TransNumLast 0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum 0
Source B	Equip.CTL.CycleMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.CycleMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CycleMsg
PerfTrans	EquipStatus04.CycleRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2509
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

Diagnostic

ADD	
Source A	CycleSeqER 1953
Source B	1
Dest	CycleSeqER 1953

Diagnostic

MOVE	
Source	LNi 27
Dest	CycleSeqER_LNi 11

Diagnostic

MOVE	
Source	EQi 26
Dest	CycleSeqER_EQi 32

Diagnostic

MOVE	
Source	CycleMsg.Header.TransNum 0
Dest	CycleSeqER_TransNum 2163

Diagnostic

MOVE	
Source	Equip.CTL.CycleMsg.TransNumNext 0
Dest	CycleSeqER_TransNumNext 2152

Receive Equipment Cycle Unsolicited Message

Load new message in Ring Buffer

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

ADD
Source A: LNi 27
Source B: EQi 26
Dest: LnEqSum 16

ADD
Source A: LnEqSum 16
Source B: CycleMsg.Data.SeqNum 0
Dest: LnEqSeqSum 27

RingBuffer Modulo Result
MOD
Source A: LnEqSeqSum 27
Source B: 2
Dest: BufferID 3

RingBuffer Modulo Result
EQ
Source A: BufferID 3
Source B: 1

UseCycleRB01

Load Equipment Cycle Message into Ring Buffer
JSR
Routine Name: _271_EquipCycleRxLoadRB01

UseCycleRB01 /

UseCycleRB02

Load Equipment Cycle Message into Ring Buffer
JSR
Routine Name: _272_EquipCycleRxLoadRB02

Receive Equipment Cycle Unsolicited Message

Prepare for next message

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source CycleMsg.Header.TransNum 0
Dest Equip.CTL.CycleMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A CycleMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.CycleMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.CycleMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.CycleMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.CycleMsg.TransNumNext 0

Receive Equipment Cycle Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

16

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CycleMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CycleMsg.TransLast
0

Equipment Data Lines

COP
Source CycleMsg
Dest EDC_EqRxDataAux.Line[LNi].Equip[EQi].Cycle
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Length CFG.CycleMSG.n
<EDC.CFG.CycleMSG.n>

Exit Routine

17

lbl_Exit
LBL

NOP

(End)

Load Equipment Cycle Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 271
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Cycle Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Cycle01.WritePtrErr <EDC.RB_Cycle01.WritePtrErr>
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest RB_Cycle01.WritePtrErrTag <EDC.RB_Cycle01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest RB_Cycle01.WritePtrErrRec <EDC.RB_Cycle01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2513
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

Load Equipment Cycle Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Cycle01.FullBlockWrite <EDC.RB_Cycle01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Cycle01.WriteBlockErr <EDC.RB_Cycle01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Cycle01.WriteBlockErrCnt <EDC.RB_Cycle01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Cycle01.WriteBlockErrCnt <EDC.RB_Cycle01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2514
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Cycle Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Cycle Message
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 1	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec01[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 2	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec02[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 3	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec03[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 4	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec04[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 5	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec05[RB_Cycle01.WriteRecPtr] Length 1

Load Equipment Cycle Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.WriteTagPtr
	<EDC.RB_Cycle01.WriteTagPtr>
	2
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.WriteRecPtr
	<EDC.RB_Cycle01.WriteRecPtr>
	547
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.ReadTagPtr
	<EDC.RB_Cycle01.ReadTagPtr>
	2
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.ReadRecPtr
	<EDC.RB_Cycle01.ReadRecPtr>
	546
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2511
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

Load Equipment Cycle Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 1
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

Load Equipment Cycle Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Cycle01.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564
Source B 1
Dest RB_Cycle01.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Cycle Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 272
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Cycle Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Cycle02.WritePtrErr <EDC.RB_Cycle02.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest RB_Cycle02.WritePtrErrTag <EDC.RB_Cycle02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest RB_Cycle02.WritePtrErrRec <EDC.RB_Cycle02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2513
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

Load Equipment Cycle Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Cycle02.FullBlockWrite <EDC.RB_Cycle02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Cycle02.WriteBlockErr <EDC.RB_Cycle02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Cycle02.WriteBlockErrCnt <EDC.RB_Cycle02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Cycle02.WriteBlockErrCnt <EDC.RB_Cycle02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2514
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Cycle Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

Equipment Data Cycle Message

COP
Source CycleMsg
Dest EDC_RB_Cycle02_Rec01[RB_Cycle02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 2

Equipment Data Cycle Message

COP
Source CycleMsg
Dest EDC_RB_Cycle02_Rec02[RB_Cycle02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 3

Equipment Data Cycle Message

COP
Source CycleMsg
Dest EDC_RB_Cycle02_Rec03[RB_Cycle02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 4

Equipment Data Cycle Message

COP
Source CycleMsg
Dest EDC_RB_Cycle02_Rec04[RB_Cycle02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 5

Equipment Data Cycle Message

COP
Source CycleMsg
Dest EDC_RB_Cycle02_Rec05[RB_Cycle02.WriteRecPtr]
Length 1

Load Equipment Cycle Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteRecPtr
<EDC.RB_Cycle02.WriteRecPtr>
437
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadRecPtr
<EDC.RB_Cycle02.ReadRecPtr>
436
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2511
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Load Equipment Cycle Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 1
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

Load Equipment Cycle Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Cycle02.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368
Source B 1
Dest RB_Cycle02.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR	
Input Par	Trans.Code
	<EDC.Trans.Code>
Input Par	Trans.Time
	<EDC.Trans.Time>
Input Par	Trans.LNi
	<EDC.Trans.LNi>
Input Par	Trans.EQi
	<EDC.Trans.EQi>
Input Par	Trans.Num
	<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	901
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[901] -1329175072
Source B	1
Dest	Prog.RoutineScan[901] -1329175072

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

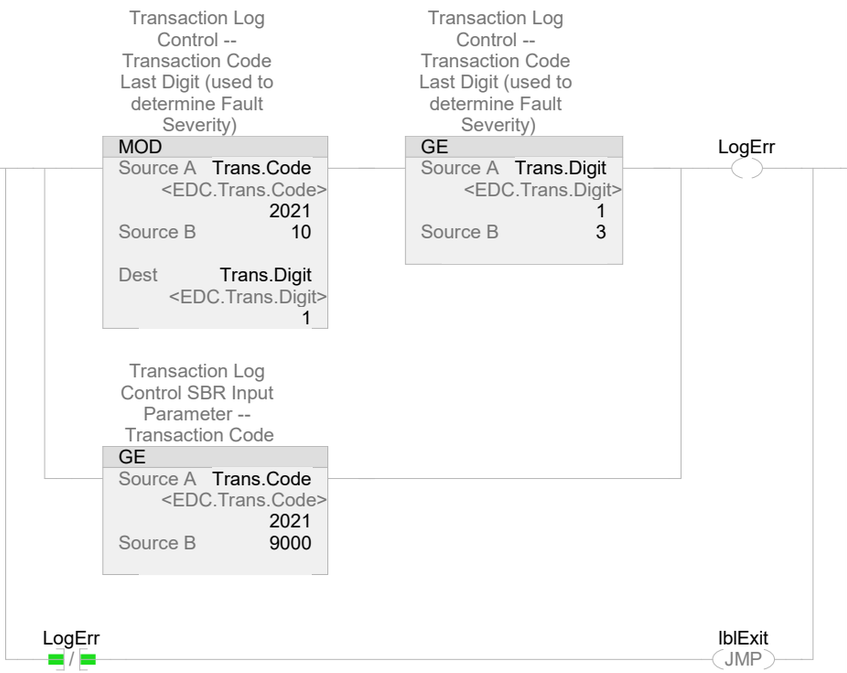
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

eAssy_EDC_Gway
MainTask
EquipRxBufferData
_201_Main
Ladder Diagram 1
_202_Fault
Ladder Diagram 5
_211_Config
Ladder Diagram 9
_220_LineLoop
Ladder Diagram 17
_225_EquipLoop
Ladder Diagram 20
_227_EquipFaultRxBulk
Ladder Diagram 28
_230_EquipFaultRx
Ladder Diagram 44
_231_EquipFaultRxLoadRB01
Ladder Diagram 61
_232_EquipFaultRxLoadRB02
Ladder Diagram 67
_233_EquipFaultRxLoadRB03
Ladder Diagram 73
_234_EquipFaultRxLoadRB04
Ladder Diagram 79
_235_EquipFaultRxLoadRB05
Ladder Diagram 85
_236_EquipFaultRxLoadRB06
Ladder Diagram 91
_240_EquipQualityRx
Ladder Diagram 97
_241_EquipQualityRxLoadRB01
Ladder Diagram 116
_242_EquipQualityRxLoadRB02
Ladder Diagram 123
_243_EquipQualityRxLoadRB03
Ladder Diagram 130
_244_EquipQualityRxLoadRB04
Ladder Diagram 137
_245_EquipQualityRxCarrierTrack
Ladder Diagram 144
_245_EquipQualityRxCarrierTrack_CfgAnim
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